LAYRATE

EGG COUNTING & FORECASTING SYSTEM

User Manual

Offline Poultry Farm Management System

Prepared for **Albert's EggVenture**
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CHAPTER 1

Introduction

About LayRate

LayRate is the farm management system of Albert's EggVenture. It keeps the farm's daily records in one place: cages and hens, eggs laid and harvested, feed, the coop environment, egg orders, and reports. It also counts eggs and reads the coop temperature and humidity through sensors in the cages, and it can forecast egg production from the farm's history.

LayRate runs on a small computer (a Raspberry Pi) at the farm. It does not need the internet. You open it in the web browser of any phone, tablet or laptop that is connected to the farm's own Wi-Fi.

Who this manual is for

This manual is for everyone who uses LayRate at the farm. It explains what each screen shows and gives step-by-step instructions for the daily work. You do not need any technical knowledge.

User roles

Every account is either an **Operator** or an **Admin**. Operators do the daily record keeping. Admins can do everything an operator can, and also set up the farm and manage accounts.

Task	Operator	Admin
View every section except Forecast	Yes	Yes
Log eggs, feeding and readings; register hens; add stock, orders and notes	Yes	Yes
Export and print reports	Yes	Yes
Add, edit, move or delete cages and save the farm layout	No	Yes
Delete records (egg, stock, order, feed and mortality records)	No	Yes
Add, edit or delete hardware; switch the cooling fan	No	Yes
Open the Forecast page and generate forecasts	No	Yes
Set egg prices, the system time, backups; manage user accounts	No	Yes

How to read this manual

- Names of buttons, tabs and fields are in **bold**, exactly as they appear on the screen.
- **+** means the round **Quick Actions** button at the bottom right of the screen. Most "add" actions start there.
- ⓘ means the small info button on a summary card. Tap it to read what the number means and how it is counted.
- The **User Manual** tab on your Profile page has this manual as a PDF and guided walkthroughs of the main tasks (also written out in Step-by-Step Tasks).

CHAPTER 2

Getting Started

What you need

- A phone, tablet or laptop with a recent web browser: Chrome, Firefox, Brave or Safari.
- The farm Wi-Fi. LayRate's computer makes its own Wi-Fi network at the farm, so no internet is needed.
- Your LayRate account (email and password). Ask an Admin if you do not have one yet.

Farm connection details (to be filled in by the Admin)

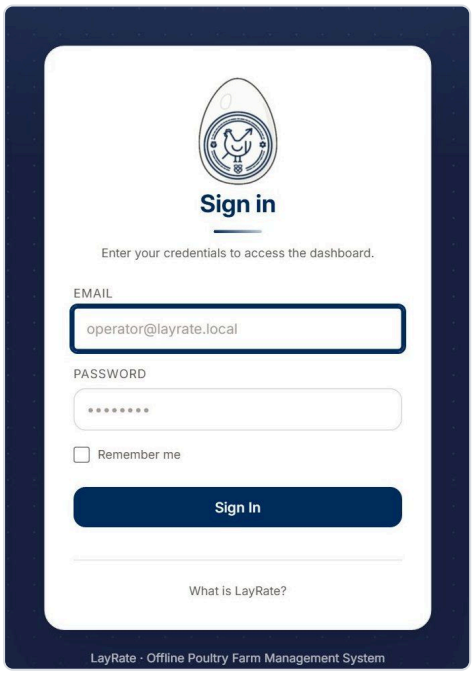
Wi-Fi name _____

Wi-Fi password _____

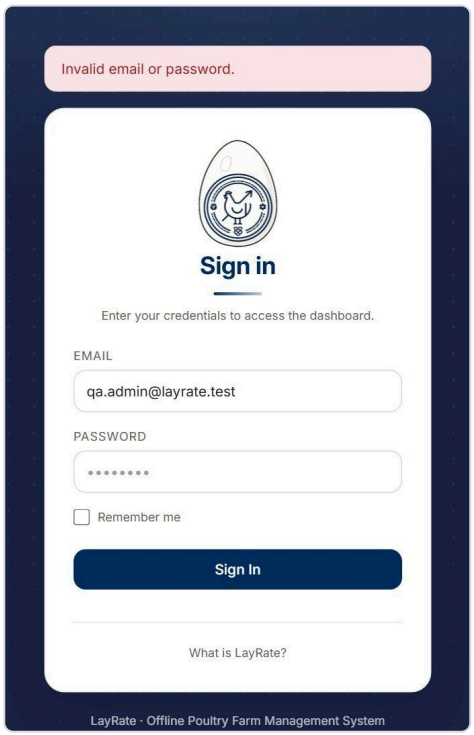
LayRate address _____

Signing in

1. Connect your device to the farm Wi-Fi.
2. Open your browser and go to the LayRate address. The **Sign in** page opens.
3. Type your **EMAIL** and **PASSWORD**.
4. Tick **Remember me** if this is your own device and you want to stay signed in.
5. Tap **Sign In** or press Enter. The Dashboard opens.



The Sign in page.

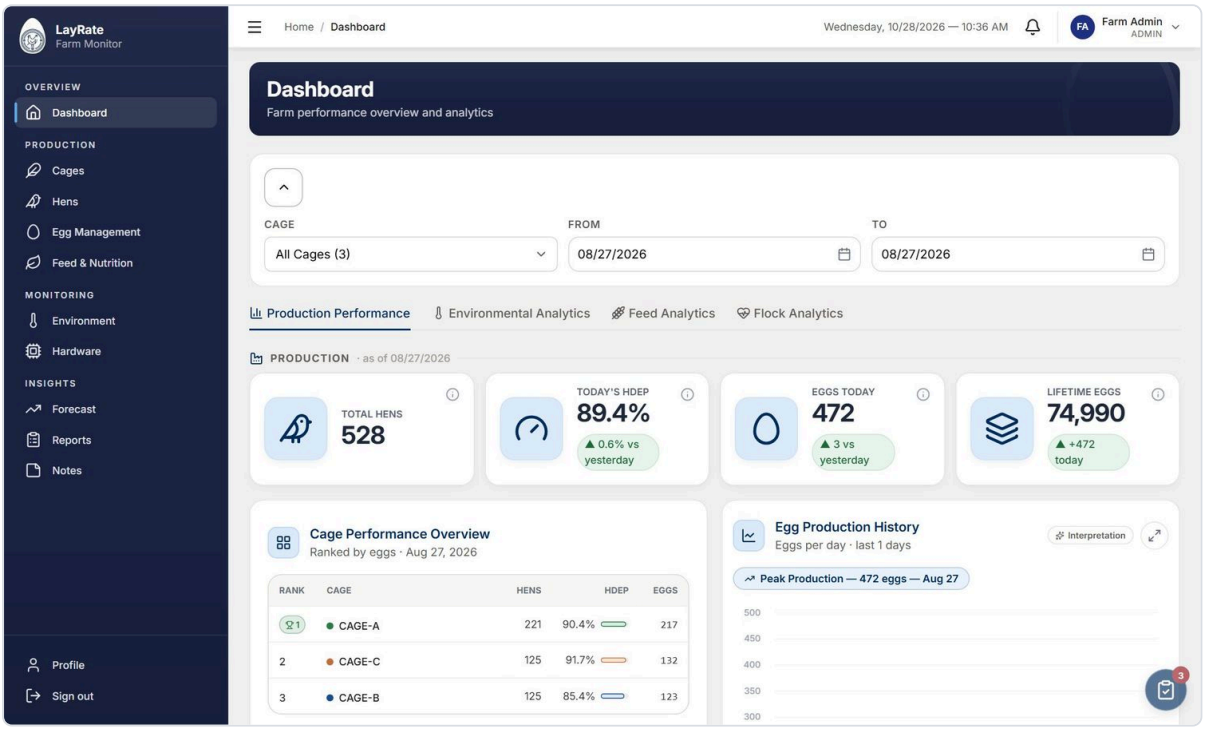


A failed sign-in. The email stays filled in, so you only retype the password.

If something is wrong, a red message appears above the form and you can try again right away:

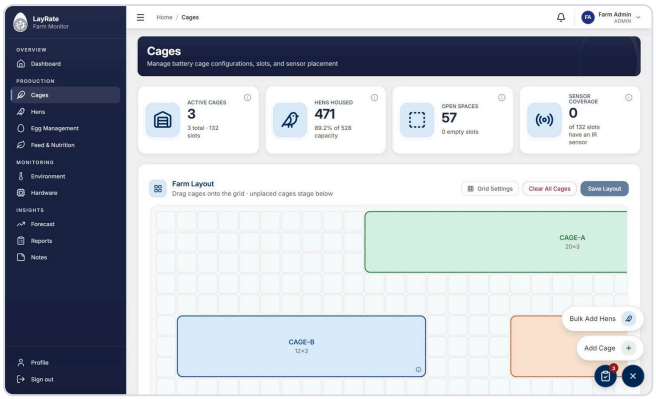
Message	What to do
Invalid email or password.	Check the spelling of your email and your password, then try again. For safety, LayRate does not say which of the two is wrong.
Too many sign-in attempts. Try again in ... seconds.	After 5 wrong tries in a row, wait for the time shown, then try again.
This account has been deactivated.	Ask an Admin to reactivate your account.
Unable to reach the sign-in service / The sign-in request timed out.	Check that your device is still on the farm Wi-Fi, then try again.

The main screen

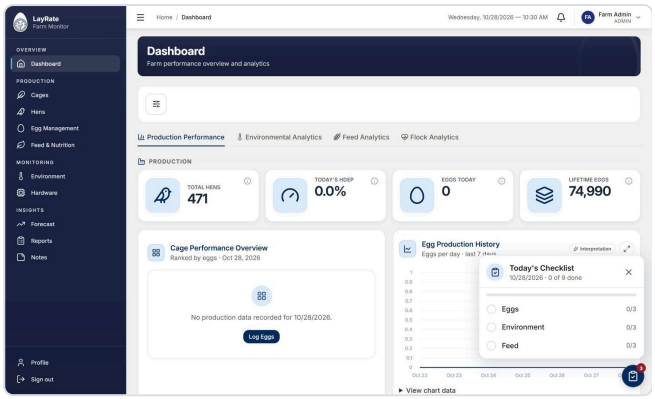


The parts of the main screen, shown on the Dashboard.

- **Sidebar (left).** Opens each section: Dashboard, Cages, Hens, Egg Management, Feed & Nutrition, Environment, Hardware, Forecast (Admins only), Reports and Notes. **Profile** and **Sign out** are at the bottom. On a phone, tap the menu button (☰) at the top left to show it.
- **Top bar.** Shows where you are, today's date and time, the **notification bell**, and your name. Tap your name for **Profile**, **Settings** and **Sign out**.
- **Summary cards.** The boxes at the top of each page with the key numbers. Tap ⓘ on a card to see what it means.
- **Tabs and filters.** Many pages have tabs (for example **Inventory** and **Mortality**) and filters such as cage and date. On a phone, tap **Show filters** to open them.
- **Quick Actions (+) and Today's Checklist.** The round buttons at the bottom right. **+** lists the add actions for the page you are on. The checklist button shows the day's record-keeping tasks; the red number is how many are still left.



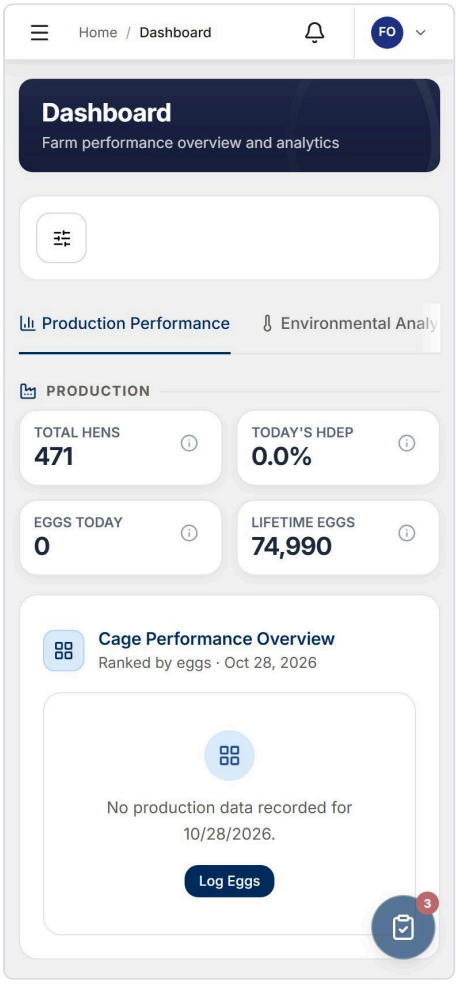
Quick Actions (+) opened on the Cages page.



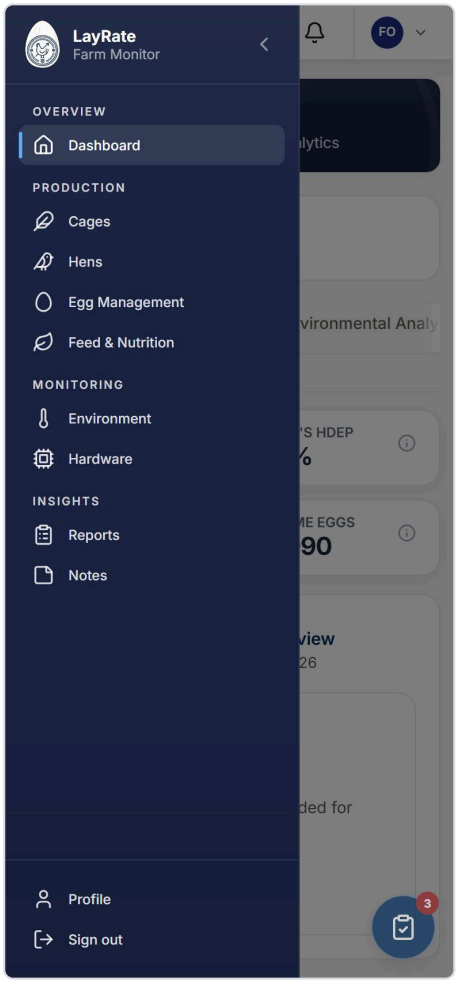
Today's Checklist.

Using LayRate on a phone

Every page works on a phone. Cards and tables stack into one column; wide tables can be swiped sideways. Open the sidebar with the menu button (☰).



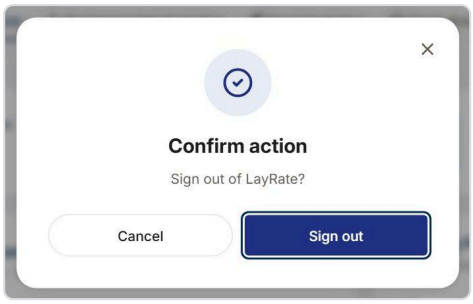
Dashboard on a phone.



The menu opened on a phone.

Signing out

1. Tap **Sign out** at the bottom of the sidebar (or in the menu under your name).
2. Tap **Sign out** again in the box that asks *Sign out of LayRate?* (**Cancel** keeps you signed in).



Confirming sign out.

After you sign out, the browser's Back button cannot show the farm pages again; it returns to the Sign in page. Always sign out on a shared device.

CHAPTER 3

Dashboard

The Dashboard is the farm's overview. It opens after you sign in.

Summary cards

Card	What it shows
Total Hens	Live hens in active cages.
Today's HDEP	Hen-day egg production: eggs collected that day divided by the hens, as a percentage. A small pill shows the change from the day before.
Eggs Today	Eggs collected that day across the active cages.
Lifetime Eggs	Every egg ever logged.

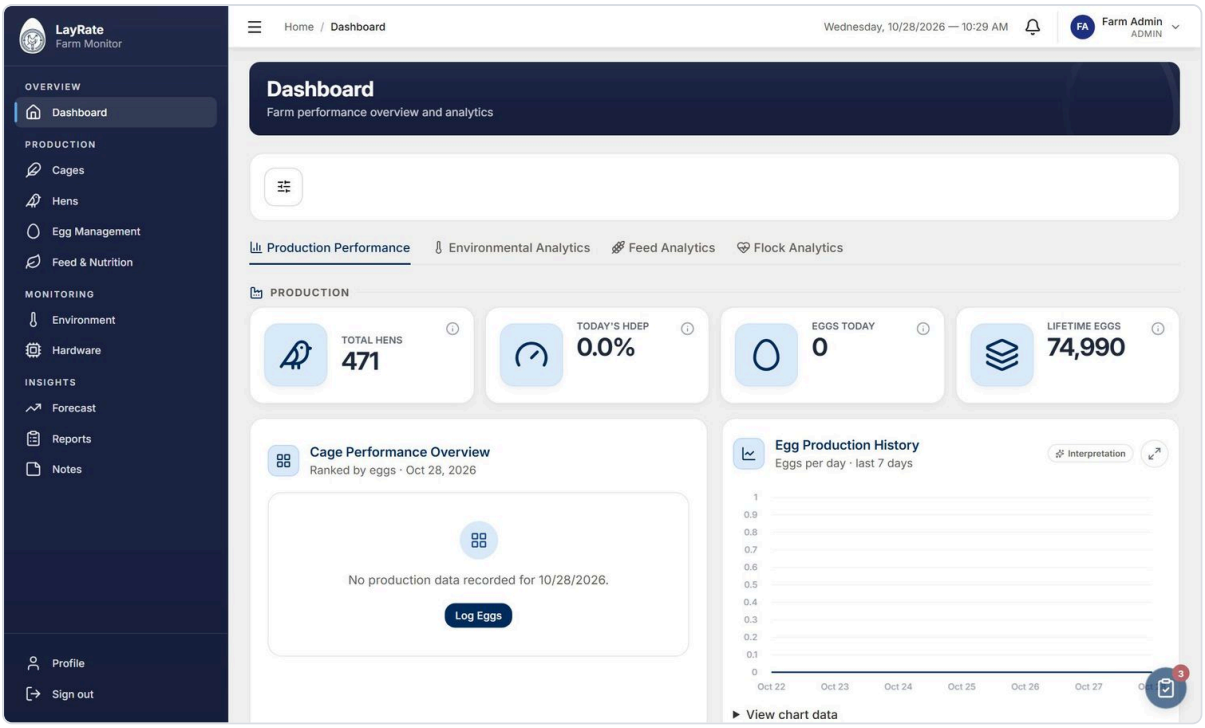
Choosing a cage and a date

Tap the filter button (the sliders icon under the page title) to show the filters:

- **CAGE**: shows the whole farm (**All Cages**) or one cage.
- **FROM** and **TO**: pick a date to see that day's numbers. Leave them empty to see today.

On a laptop the page updates as soon as you change a filter. On a phone the filters open in a panel: tap **Apply** to use them, **Cancel** to discard the changes, or **Clear** to reset them.

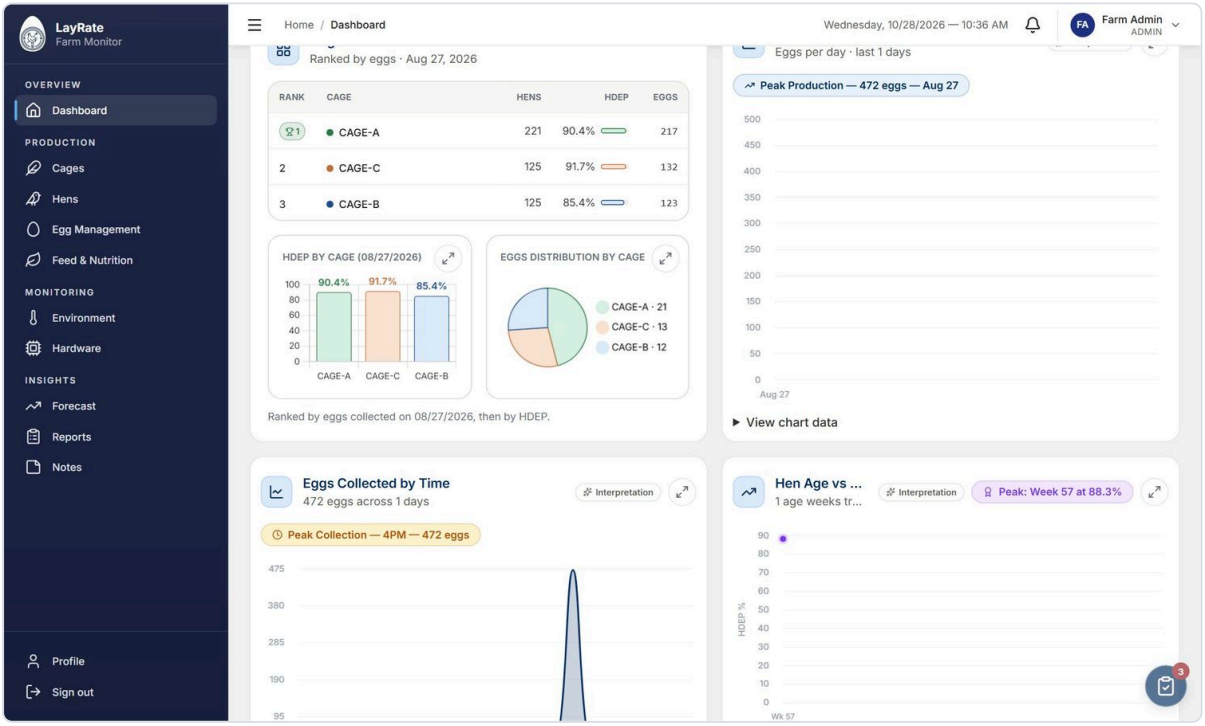
When you pick a past day, every card and the **Cage Performance Overview** show that day as it was recorded, including the number of hens on that day. A notice may say *Day Production Complete*: that day has ended and its records are final.



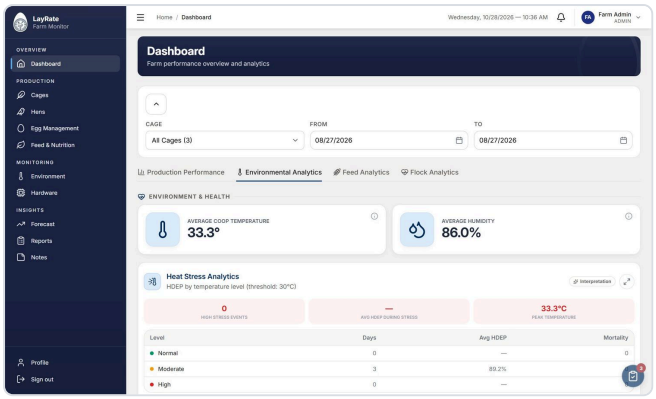
Today, before any eggs are logged. The Cage Performance Overview offers a **Log Eggs** button.

Dashboard tabs

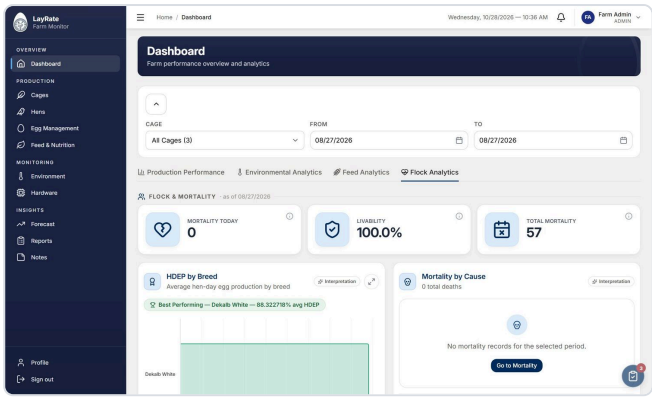
- **Production Performance**: cage ranking for the day (**Cage Performance Overview**), **Egg Production History** and other production charts.
- **Environmental Analytics**: temperature and humidity, and how they relate to production.
- **Feed Analytics**: feed used and how it compares with eggs produced.
- **Flock Analytics**: hen age, mortality and breed results.



Cage Performance Overview and Egg Production History for a chosen day.



Environmental Analytics.



Flock Analytics.

Reading the charts

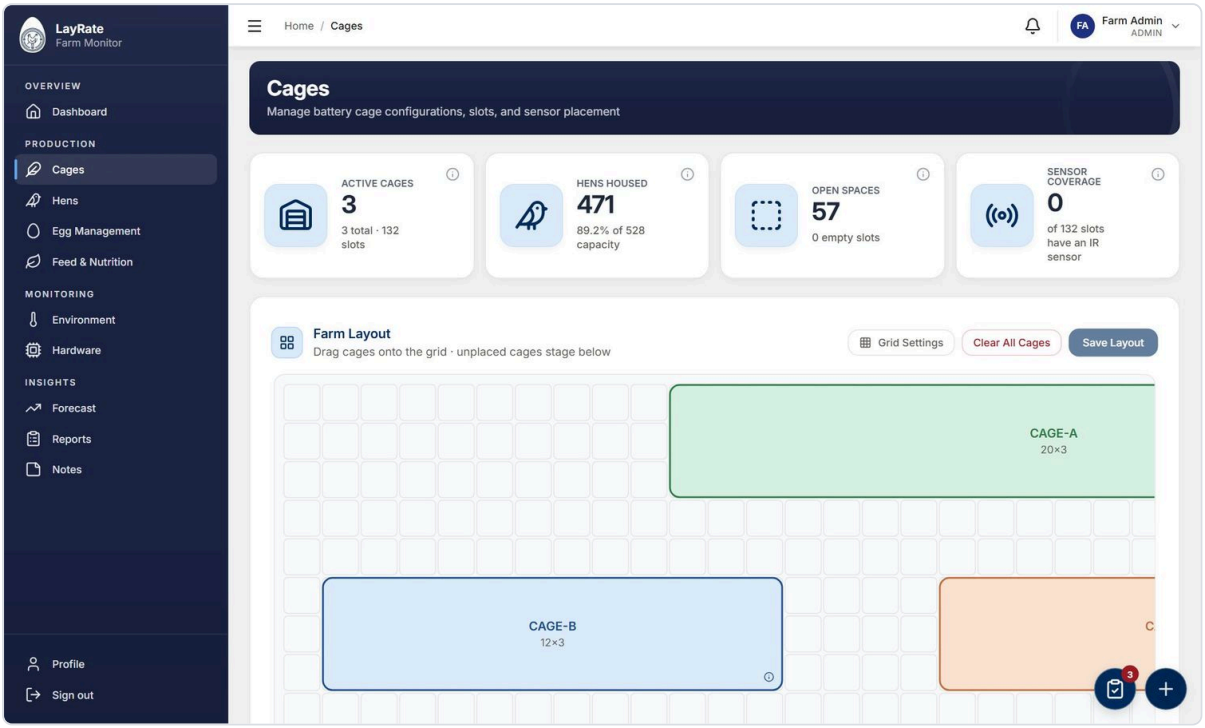
- Hover over or tap a point to see its exact value.
- Tap the full-screen button (⤢) on a chart to open it large.
- **Interpretation** explains in plain words what the chart shows.
- **View chart data** under a chart shows the same numbers as a table.

> **Note** When you first open the Dashboard each day, a **Yesterday's Production Record** box shows yesterday's eggs, HDEP, mortality and feed. Tap **Got it** to close it.

CHAPTER 4

Cages

The Cages page shows the **Farm Layout**: a map of the coop with every cage in its place. Each cage is split into rows and **slots**; a slot is one compartment that holds a few hens.

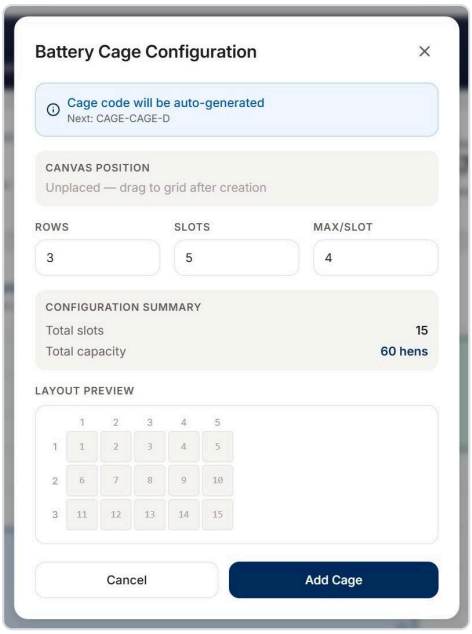


The Cages page with the Farm Layout.

Summary cards: **Active Cages**, **Hens Housed** (hens placed out of total space), **Open Spaces** and **Sensor Coverage** (slots with an egg-counting sensor).

Adding a cage (Admin)

1. Tap **+**, then **Add Cage**.
2. Set **Rows**, **Slots** (per row) and **Max/Slot** (the most hens one slot can hold).
3. Tap **Add Cage**. The new cage appears under **Unplaced Cages**.
4. Drag the cage into the Farm Layout grid. On a touchscreen, turn on **Drag cages** first.
5. Tap **Save Layout**.

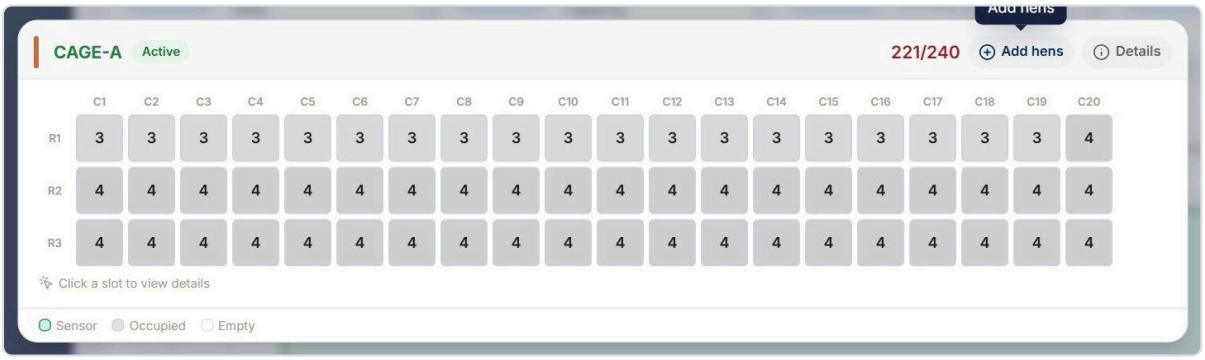


Add Cage.

Cage details and actions

Tap the **info** button (ⓘ) at the corner of a cage. A panel shows every slot with the number of hens in it. Use **Add hens** to place hens in the cage, or **Details** for the full cage details and the cage actions: **Edit**, **Renumber**, **Print** and **Delete** (Admin).

Grid Settings changes the size of the layout grid. **Clear All Cages** takes every cage off the layout so you can arrange them again; tap **Save Layout** to keep any change.



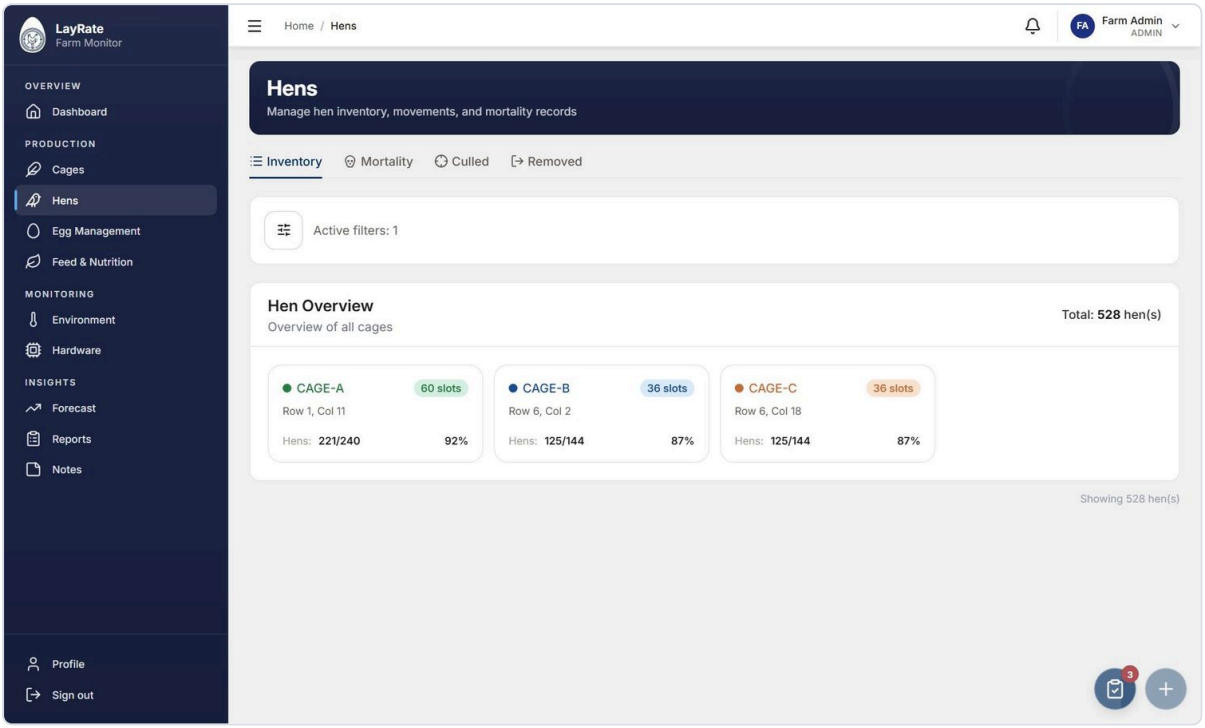
Cage details.

Bulk Add Hens (under **+**) places many hens into a cage at once.

CHAPTER 5

Hens

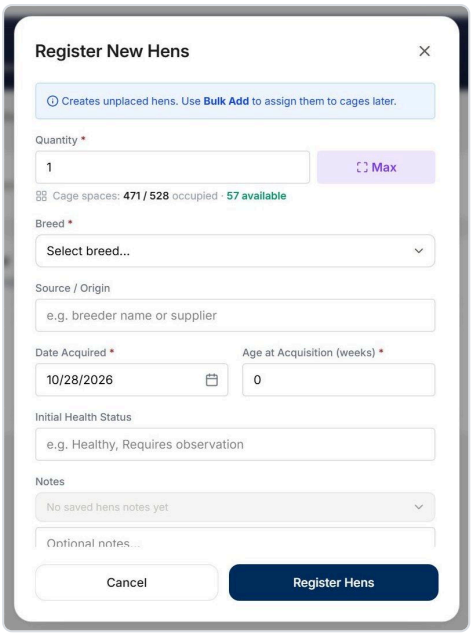
The Hens page is the flock register. Its tabs are **Inventory** (live hens), **Mortality** (deaths), **Culled** and **Removed**.



Hens: Inventory.

Registering new hens

1. Tap **+**, then **Register New Hens**.
2. Enter the **Quantity** (**Max** fills in the free cage space), **Breed**, **Date Acquired** and **Age at Acquisition (weeks)**. **Source / Origin**, **Initial Health Status** and **Notes** are optional.
3. Tap **Register Hens**. The new hens are **Unplaced**: they are not in a cage yet.
4. Open the **Unplaced** list and tap **Place into cage**, or use **Bulk Add Hens** on the Cages page, to put them in a cage.



Register New Hens.

Print Foot Tags (under **+**) prints ID tags for the hens.

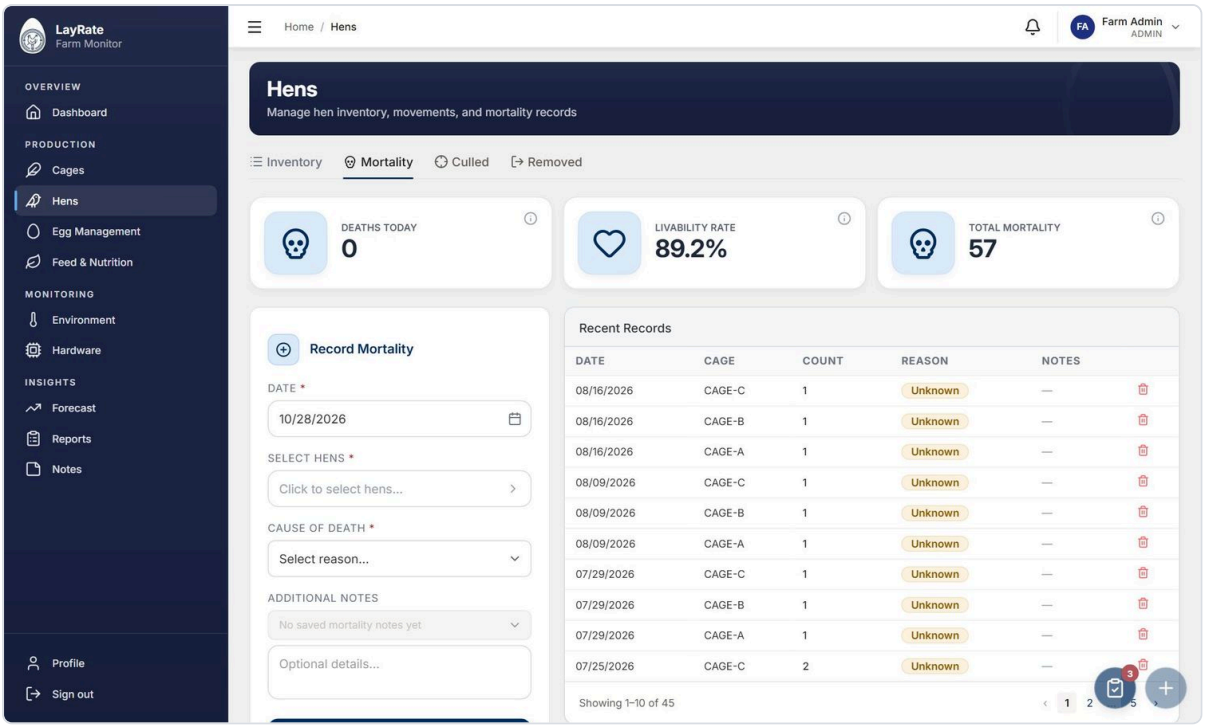
Recording what happens to a hen

Each hen (or slot) in the Inventory has action buttons:

Action	Use it to
Move hen / **Move all**	Move hens to another cage or slot.
Log health event	Record a health event: **Sick**, **Treated** or **Recovered**, with a description.
Record weight	Save a weight check.
Cull hen	Record a cull with its date and reason: Low Production, Illness, Aggression, Age or Other.
Remove hen / **Remove all**	Take hens out of the flock (for example sold or given away). Tick **Record as mortality** and choose a reason when a hen has died.

Recording deaths

1. Open the **Mortality** tab. The cards show **Deaths Today**, **Livability Rate** and **Total Mortality**.
2. In **Record Mortality**, set the **Date**, tap **Select Hens** to choose the hens that died, and choose the **Cause of Death**. Add notes if needed.
3. Save. The record appears under **Recent Records**.



The Mortality tab.

CHAPTER 6

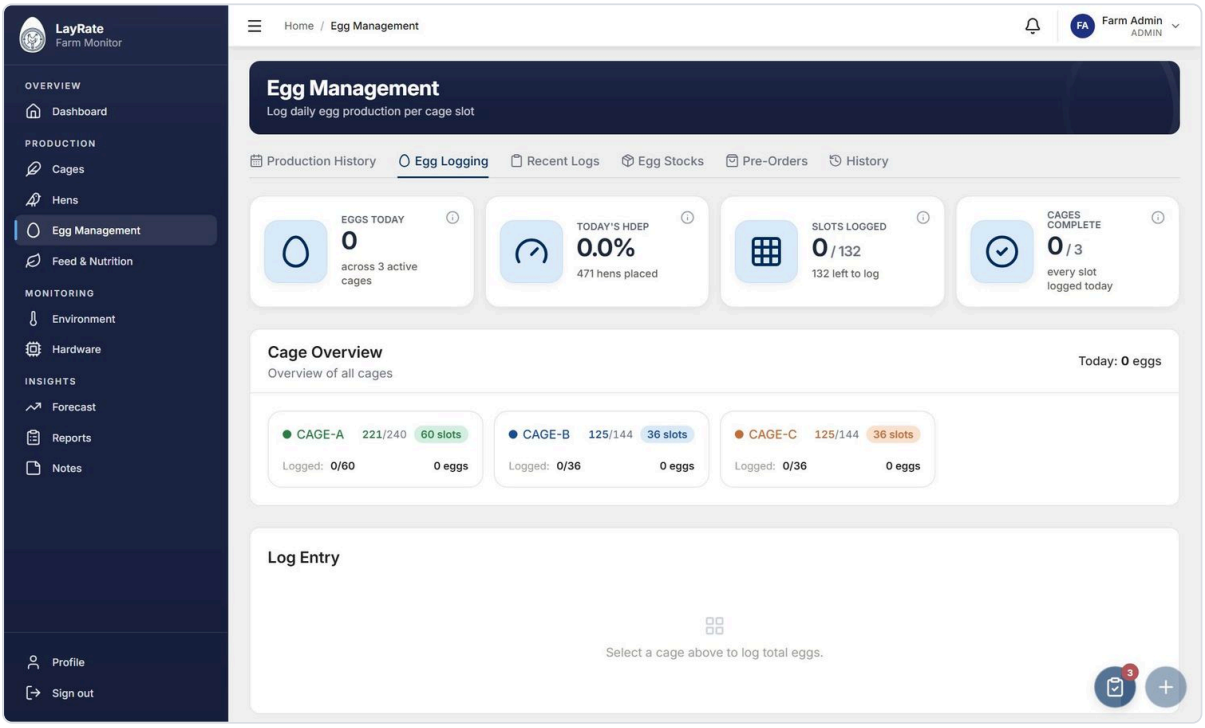
Egg Management

Egg Management has six tabs at the top:

Production History	A month calendar of the daily egg logs, with the month total, daily average, best day and days logged.
Egg Logging	Record today's eggs, cage by cage.
Recent Logs	Every egg record, with editing.
Egg Stocks	Harvested eggs on hand, by size and freshness.
Pre-Orders	Customer orders.
History	The full timeline since day one: eggs by cage, by size, and by day, week or month.

Egg Logging

Use this page every day to record the eggs laid. The **Cage Overview** shows each cage as a card with its hens, slots logged and eggs today. The summary cards show **Eggs Today**, **Today's HDEP**, **Slots Logged** and **Cages Complete**.



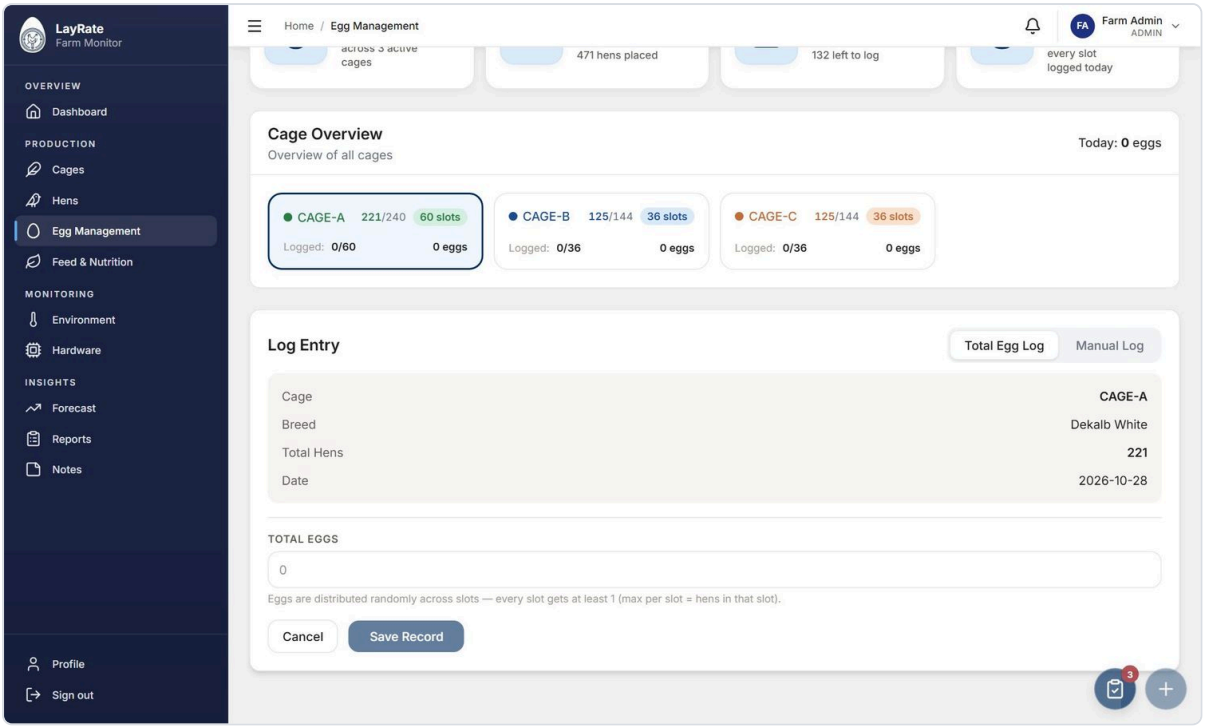
Egg Logging: Cage Overview.

Tap a cage card to open **Log Entry**. It has two ways to log:

Total Egg Log (one number for the whole cage)

1. Check the cage, breed, total hens and date shown.
2. Enter **TOTAL EGGS** for the cage. LayRate spreads the total across the cage's slots (no slot gets more eggs than it has hens).

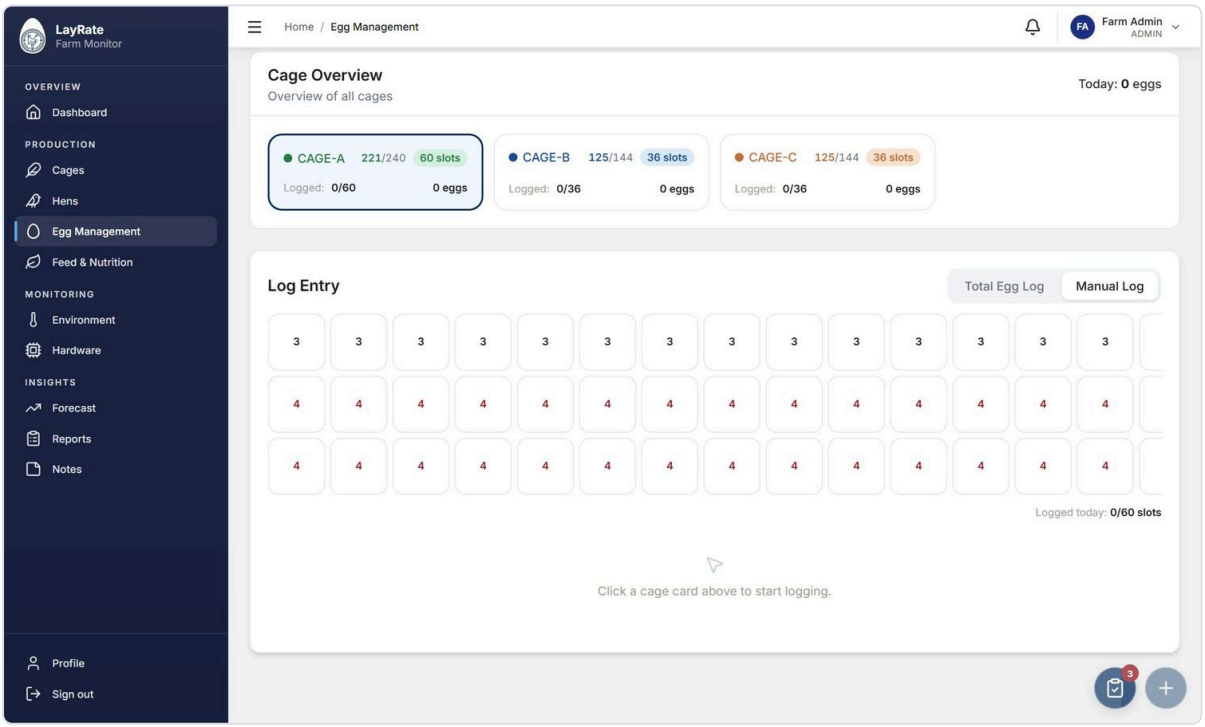
3. Tap **Save Record**.



Log Entry: Total Egg Log.

Manual Log (slot by slot)

1. Tap **Manual Log**. The cage's slots appear; the number in each slot is its hens.
2. Tap a slot to select it. To give several slots the same count, drag across them.
3. Enter the number of eggs. **0** is a valid count.
4. Tap **Save Record**. If you selected several slots, check the totals and tap **Confirm Save**.

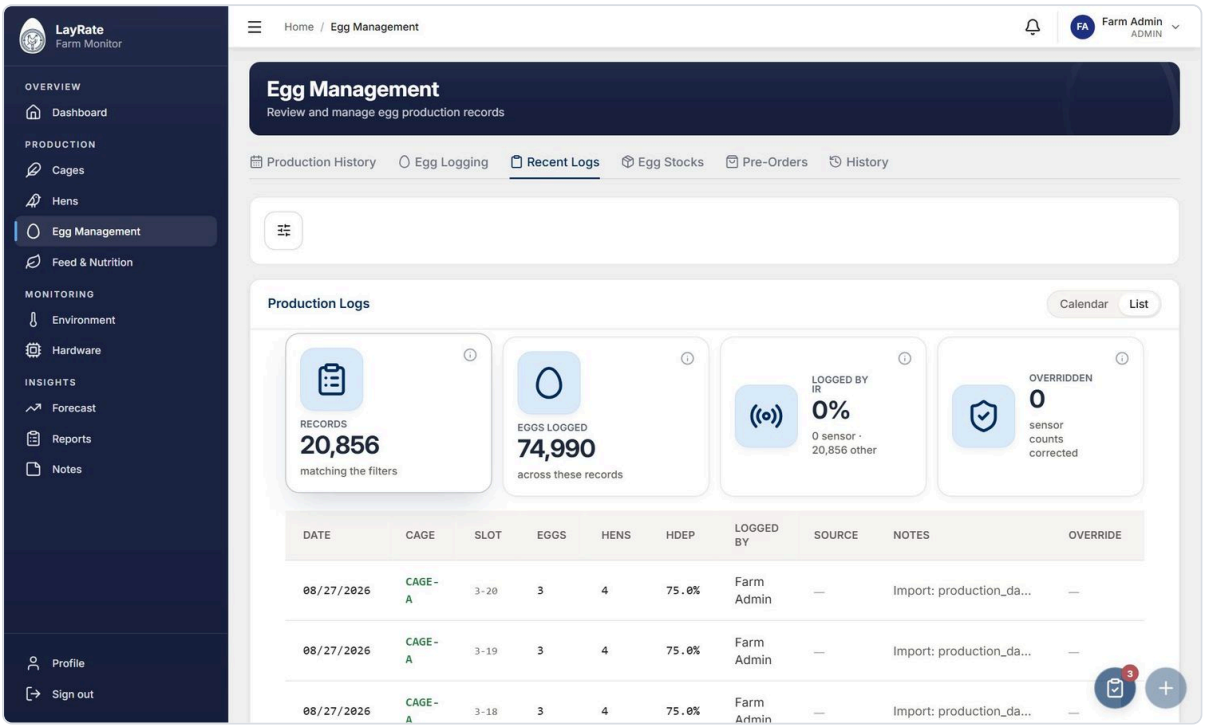


Log Entry: Manual Log.

> **Sensor slots** Slots with an IR egg-counting sensor fill in their count by themselves. If a count is wrong, use **Sensor reading – click to override**, confirm with your override PIN or password, and enter the right number. Overrides are marked in Recent Logs.

Recent Logs

Recent Logs lists every egg record with its date, cage, slot, eggs, hens, HDEP, who logged it, the source (manual or sensor) and any override. Switch between **List** and **Calendar**, and use the filters to narrow the list.

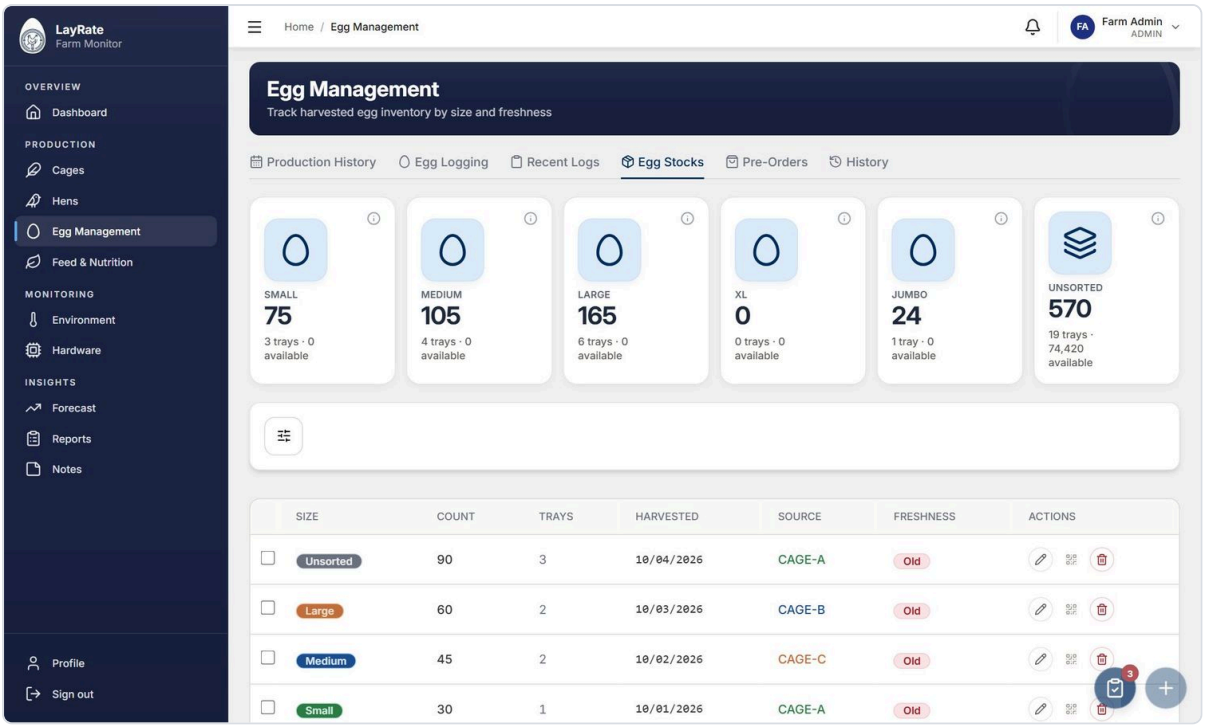


Recent Logs.

Row buttons: **Edit log**, **Reset count**, and **Delete log** (Admin).

Egg Stocks

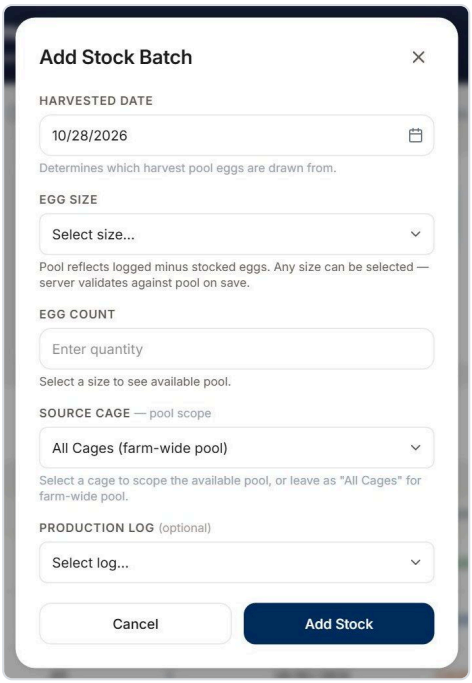
Egg Stocks tracks the harvested eggs on hand by size: Small, Medium, Large, XL, Jumbo and Unsorted. Each stock batch shows its count, trays, harvest date, source and freshness.



Egg Stocks.

- **Add Stock** (under **+**): record harvested eggs. Set the **Harvested Date**, **Egg Size** and **Egg Count**, and optionally the **Source Cage** and **Production Log**. LayRate checks the count against the eggs logged and not yet stocked.
- **Print All QR Labels** and **View QR code**: print labels for the trays.
- **Scan Egg QR for Freshness**: scan a tray label with the device camera to see how old the eggs are.
- **Egg Weights** and **Thresholds**: set the weight for each size and the low-stock warning levels.
- **Egg Prices** (Admin): set the price per tray and per piece for each size.

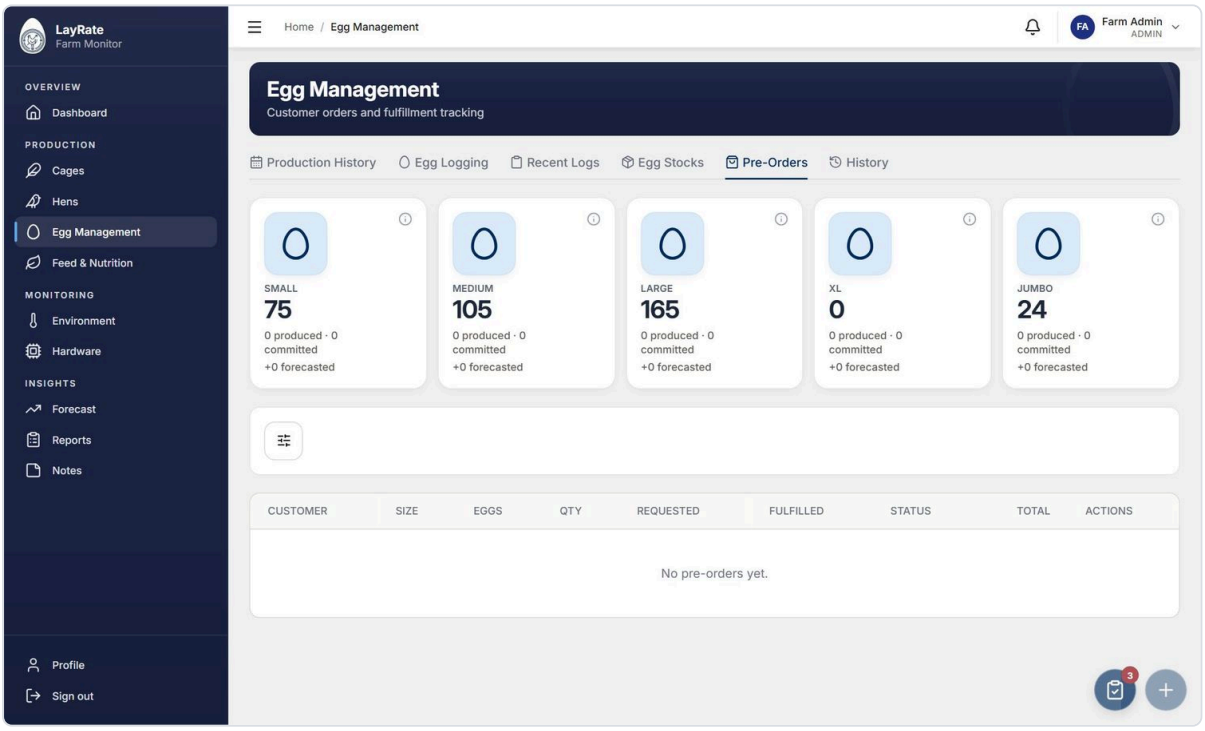
Freshness	Meaning (default setting)
Fresh	Harvested 7 days ago or less.
Aging	Harvested 8 to 14 days ago.
Old	Harvested more than 14 days ago.



Add Stock.

Pre-Orders

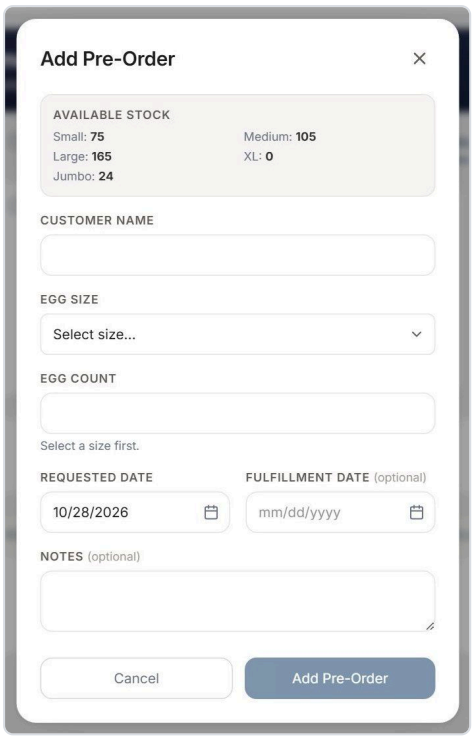
Pre-Orders records customer orders. The cards at the top show how many eggs of each size are still available after open orders.



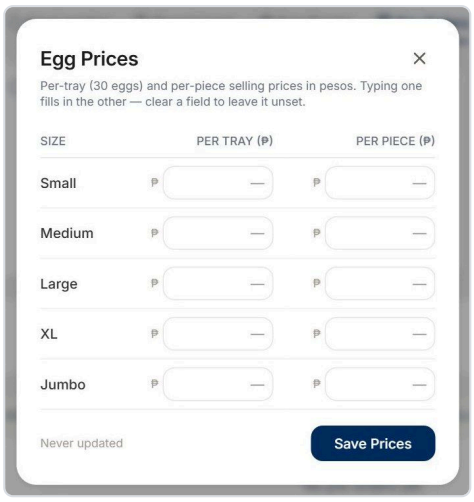
Pre-Orders.

1. Tap **+**, then **Add Pre-Order**.
2. Enter the **Customer Name**, **Egg Size**, **Egg Count** and **Requested Date**. **Fulfillment Date** and **Notes** are optional. The box at the top shows the stock available for each size.
3. Tap **Add Pre-Order**. The order starts as **Pending**, and its total is worked out from the egg prices.

4. When the eggs are handed over, edit the order and set it to **Fulfilled**; set it to **Cancelled** if the customer cancels. Mark the order as paid when payment is received.



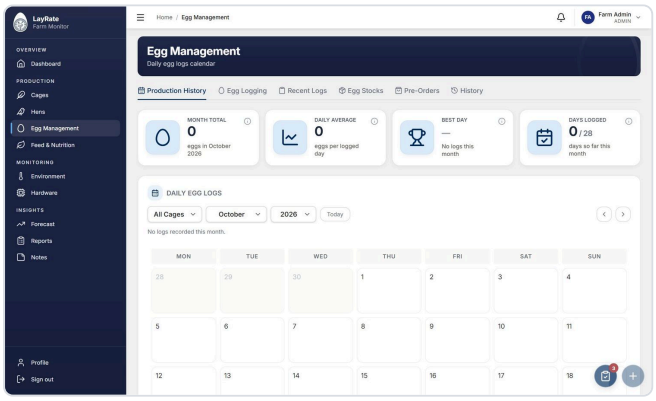
Add Pre-Order.



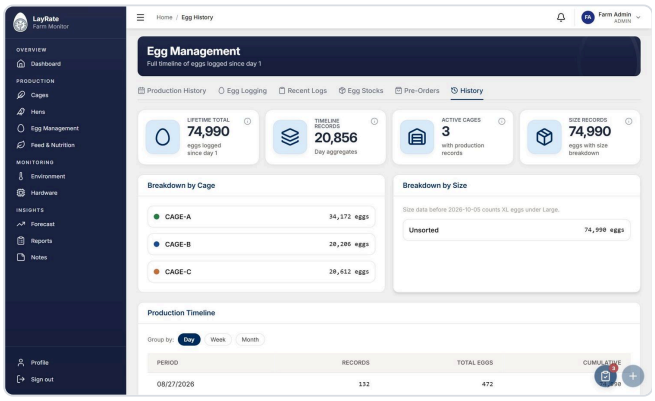
Egg Prices: per tray (30 eggs) and per piece, in pesos. Typing one fills in the other.

Production History and History

Production History shows a calendar of egg logs for one month; choose the cage, month and year at the top. **History** shows the totals since day one, broken down by cage and by size, with a **Production Timeline** you can group by **Day**, **Week** or **Month**.



Production History.



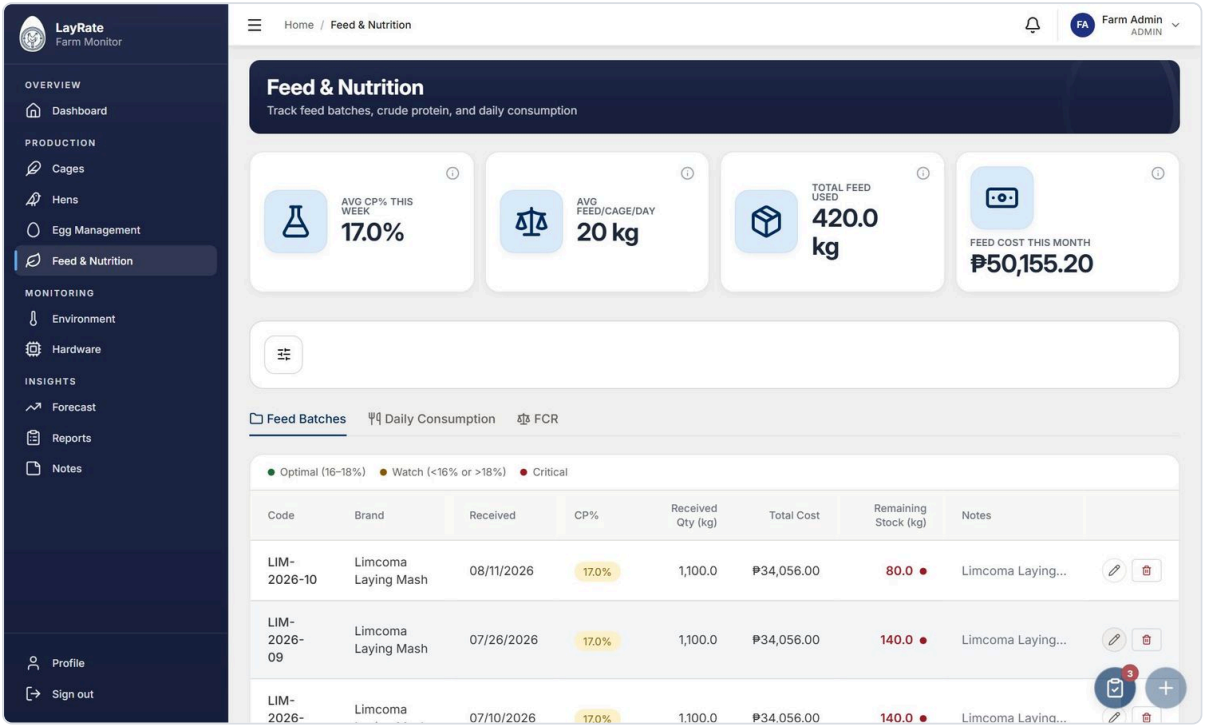
History.

CHAPTER 7

Feed & Nutrition

This page tracks feed stock and feeding. The cards show the average crude protein (CP%), average feed per cage per day, total feed used in the last 7 days and the feed cost this month. The tabs are **Feed Batches**, **Daily Consumption** and **FCR**.

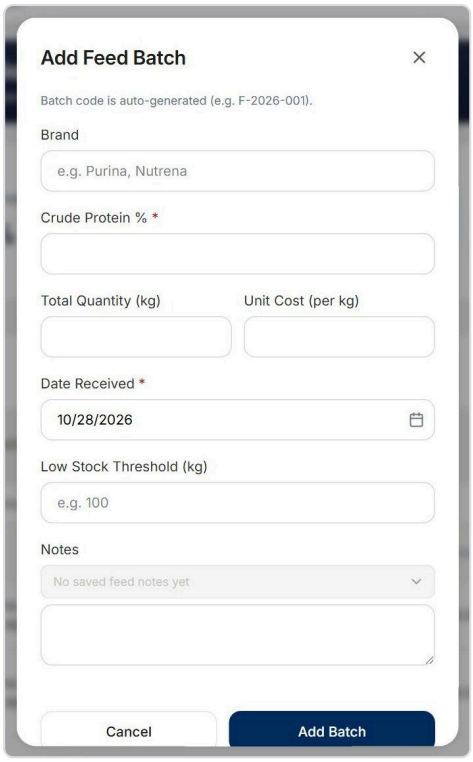
Feed Batches (feed stock)



Feed Batches.

1. Tap **+**, then **Add Feed Batch**.
2. Fill in **Brand**, **Crude Protein %** (from the feed label), **Total Quantity (kg)**, **Unit Cost** (price per kilogram), **Date Received**, **Low Stock Threshold** and optional **Notes**. The batch code is made automatically.
3. Tap **Add Batch**. **Remaining Stock (kg)** shows the kilograms left after the feeding you log.

The colored dot next to **CP%** shows the protein level: **Optimal** (16–18%), **Watch** (below 16% or above 18%) or **Critical**.

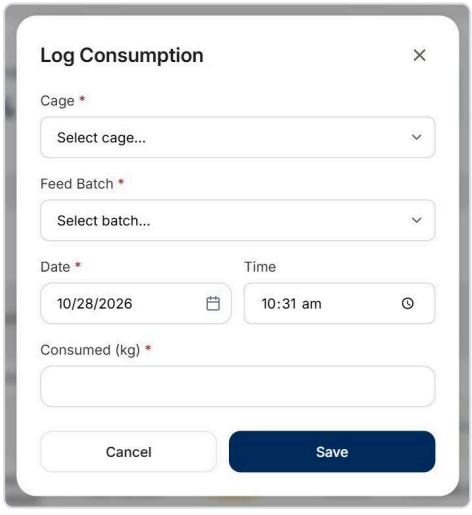


Add Feed Batch.

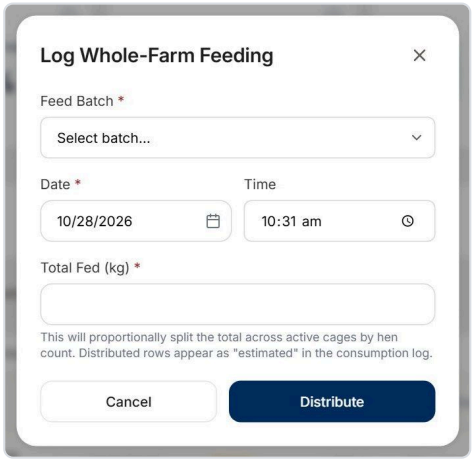
Logging feeding

Under **+** there are two ways to log feeding:

- **Add Consumption**: the feed given to one cage. Choose the **Cage** and **Feed Batch**, set the **Date** and **Time**, enter **Consumed (kg)**, and tap **Save**.
- **Log Whole-Farm Feeding**: the total feed given to the whole farm. Choose the **Feed Batch**, set the **Date** and **Time**, enter **Total Fed (kg)**, and tap **Distribute**. LayRate divides it among the active cages by their number of hens. These rows are marked **Estimated**.

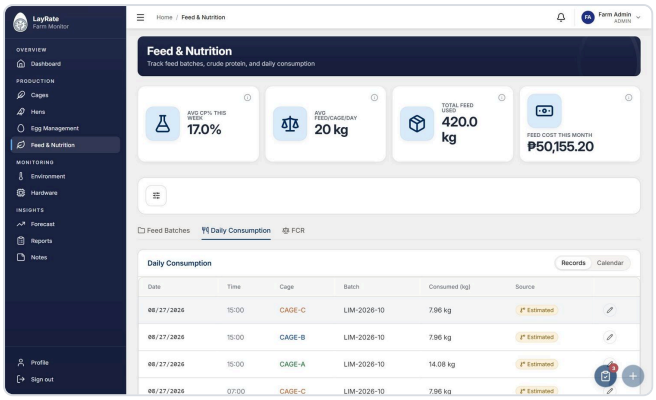


Log Consumption (one cage).

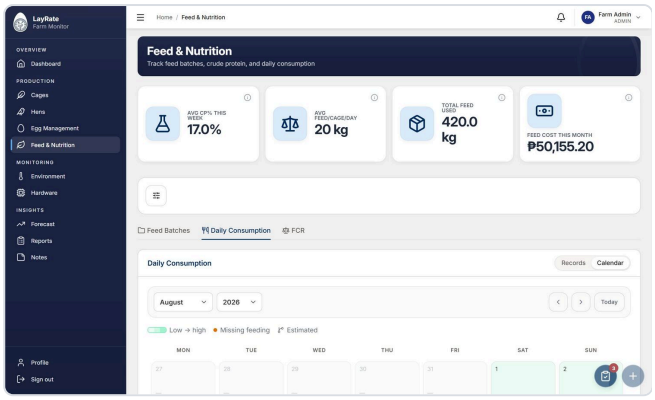


Log Whole-Farm Feeding.

The **Daily Consumption** tab lists the feeding records. Switch between **Records** and **Calendar**; on the calendar, tap a day to see its feedings.



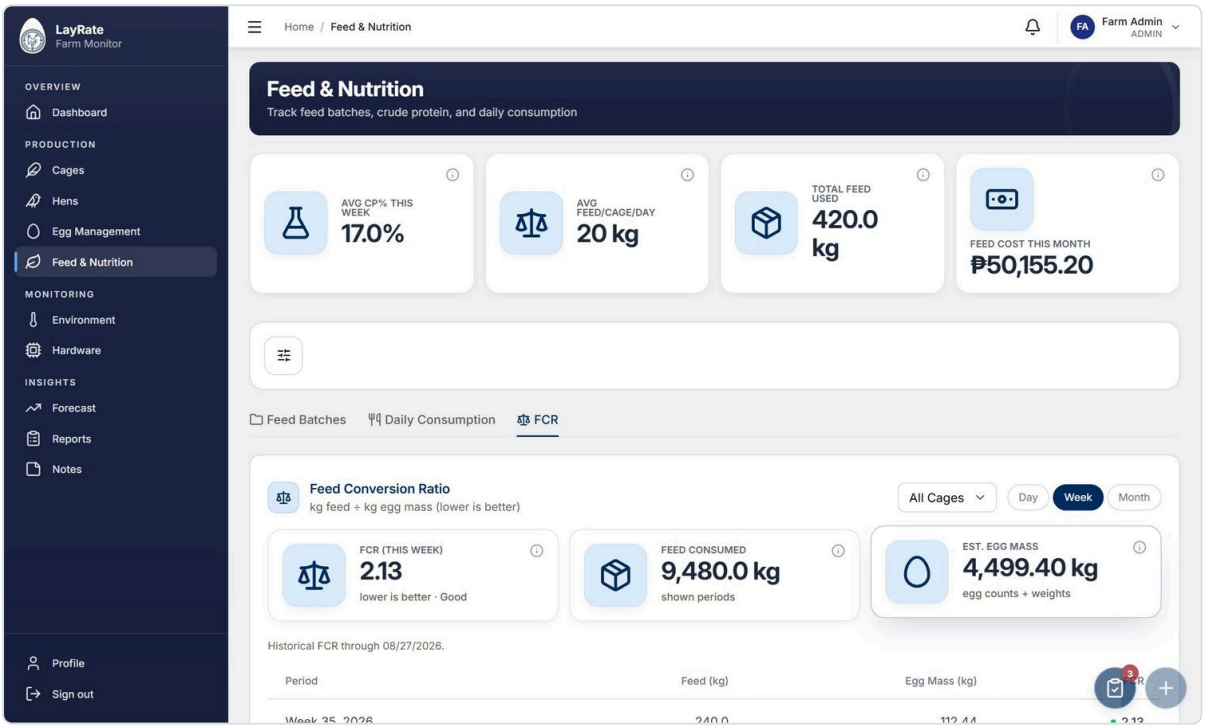
Daily Consumption: Records.



Daily Consumption: Calendar.

FCR (Feed Conversion Ratio)

FCR is the kilograms of feed used for each kilogram of eggs produced. **Lower is better.** Under 2.5 is good; above 4 needs attention. Choose a cage (or **All Cages**) and group the table by **Day**, **Week** or **Month**.



The FCR tab.

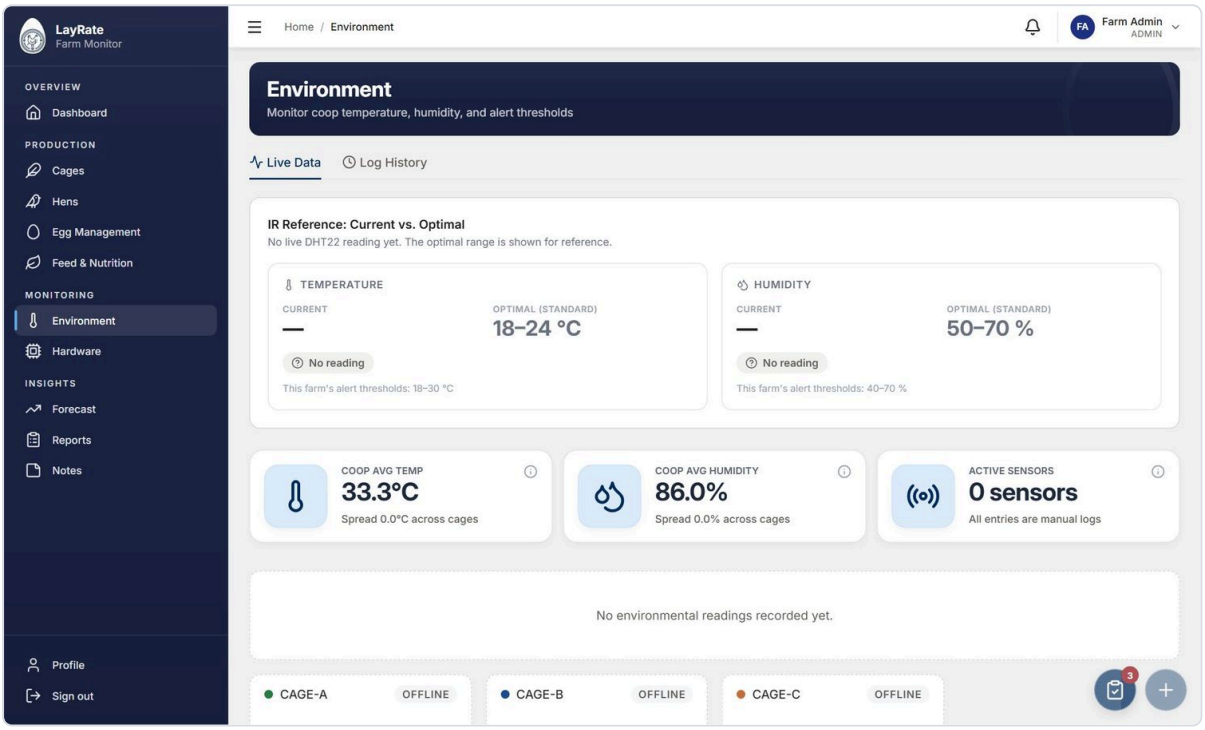
CHAPTER 8

Environment

The Environment page shows the coop's temperature and humidity from the DHT22 sensors. The **Live Data** tab has:

- **IR Reference: Current vs. Optimal**: the current reading next to the recommended range (18–24 °C and 50–70% humidity) and the farm's own alert thresholds.
- **Coop Avg Temp**, **Coop Avg Humidity** and **Active Sensors**.
- The trend charts, and a chip for each cage showing whether its sensor is online.

The **Log History** tab lists the daily averages per cage, with **Filter** and **Reset** at the top and an **Override** button on each row to correct a day's reading.

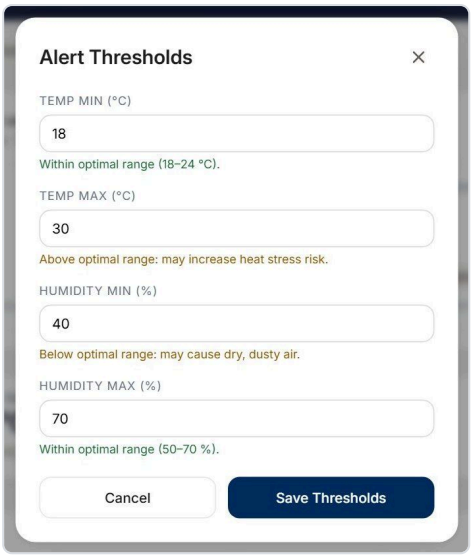


Environment: Live Data.

Each reading gets a status from your thresholds: **OK** (inside the range), **Watch** (at the edge of the range) or **Alert** (outside the range). Alerts also appear in Notifications.

Setting the alert ranges

1. Tap **+**, then **Configure Thresholds**. The **Alert Thresholds** box opens.
2. Set **Temp Min** and **Temp Max (°C)** (0 to 50 °C) and **Humidity Min** and **Humidity Max (%)** (0 to 100%). The minimum cannot be higher than the maximum. The hint under each field compares your value with the recommended range.
3. Tap **Save Thresholds**.

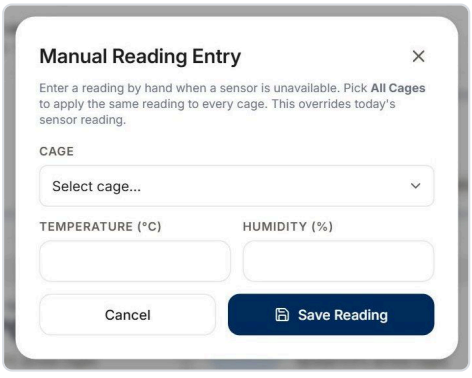


Alert Thresholds.

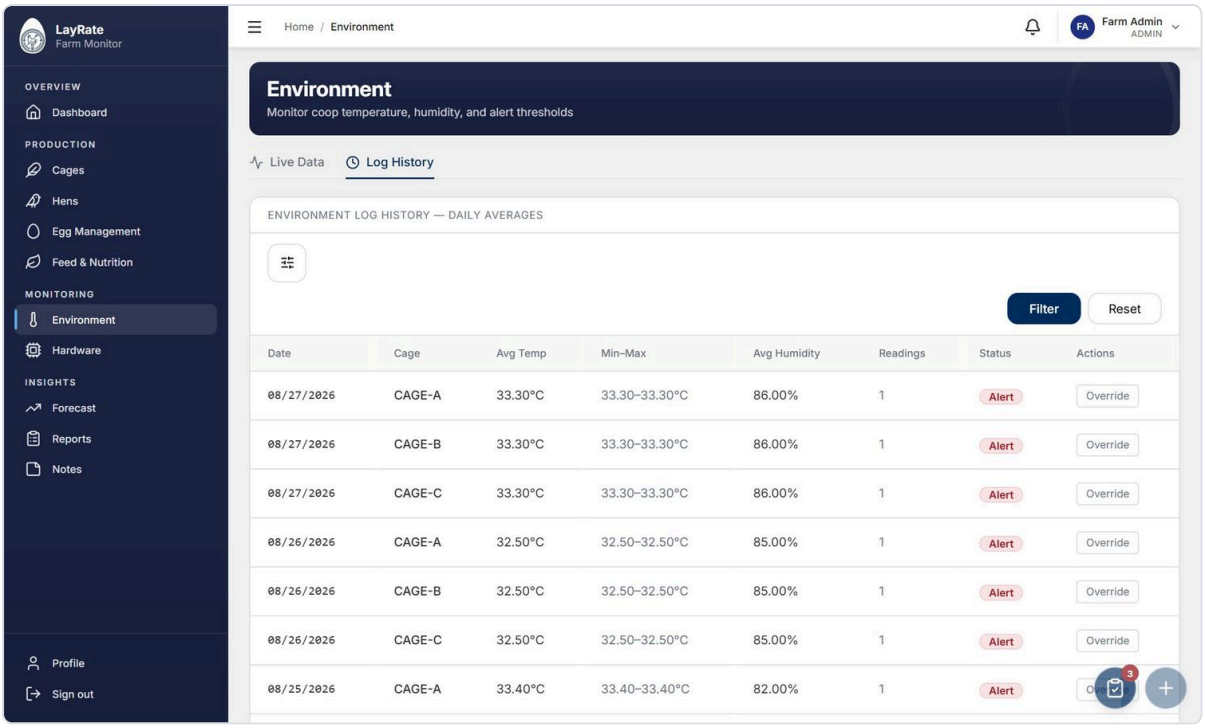
Entering a reading by hand

Use this when a sensor is not working.

1. Tap **+**, then **Manual Entry**. The **Manual Reading Entry** box opens.
2. Choose a **Cage**, or **All Cages** for the same reading everywhere.
3. Enter **Temperature (°C)** (−10 to 60) and **Humidity (%)** (0 to 100).
4. Tap **Save Reading**. It counts for today, replaces any earlier manual reading for that cage today, and takes priority over the sensor average.



Manual Reading Entry.



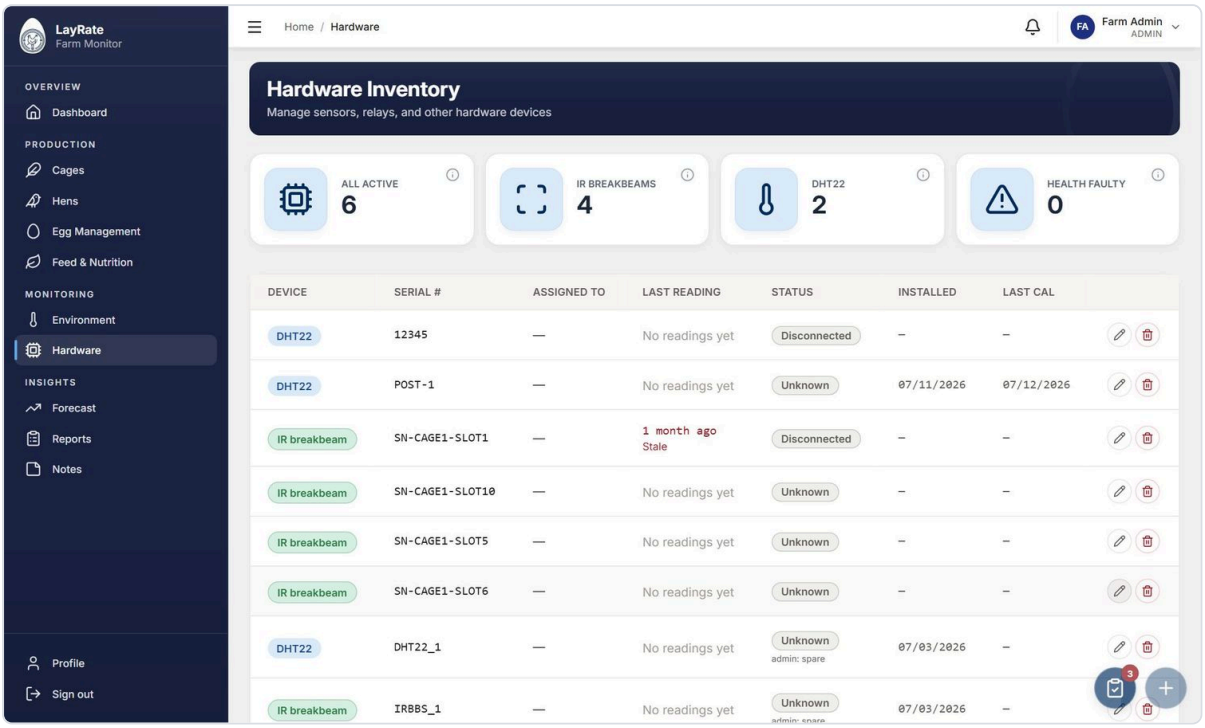
Log History.

Important **Cooling Fan** (under **+**, Admin only) switches the real fan in the coop on or off. Use it only when you mean to.

CHAPTER 9

Hardware

The Hardware Inventory lists the farm's devices: the IR breakbeam egg counters, the DHT22 temperature and humidity sensors, and the relay. For each device it shows the serial number, what it is assigned to, its last reading, its status, and its installation and last calibration dates. The cards count **All Active**, **IR Breakbeams**, **DHT22** and **Health Faulty** devices.



Hardware Inventory.

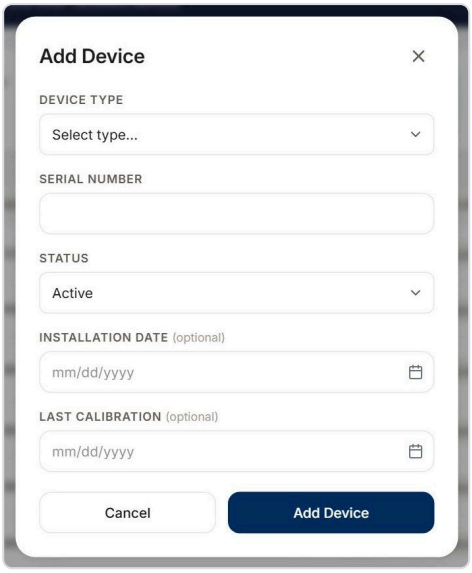
The **STATUS** column shows whether LayRate is hearing from the device (for example **Disconnected** or **Unknown** when it is not sending readings). When a device is registered, it is given one of these statuses:

Status	Meaning
Active	In use.
Spare	Kept in reserve, not assigned.
Faulty	Not working; check it.
Removed	No longer in use; kept for history.

Registering a device (Admin)

1. Tap **+**, then **Add Device**.
2. Choose the **Device Type**, enter the **Serial Number**, choose the **Status**, and add the **Installation Date** and **Last Calibration** if you know them. Depending on the type, also choose the cage or slot it serves.
3. Tap **Add Device**, then **Add Item** to confirm.

Registering a device only records it. The device must also be connected to the farm's sensor system.

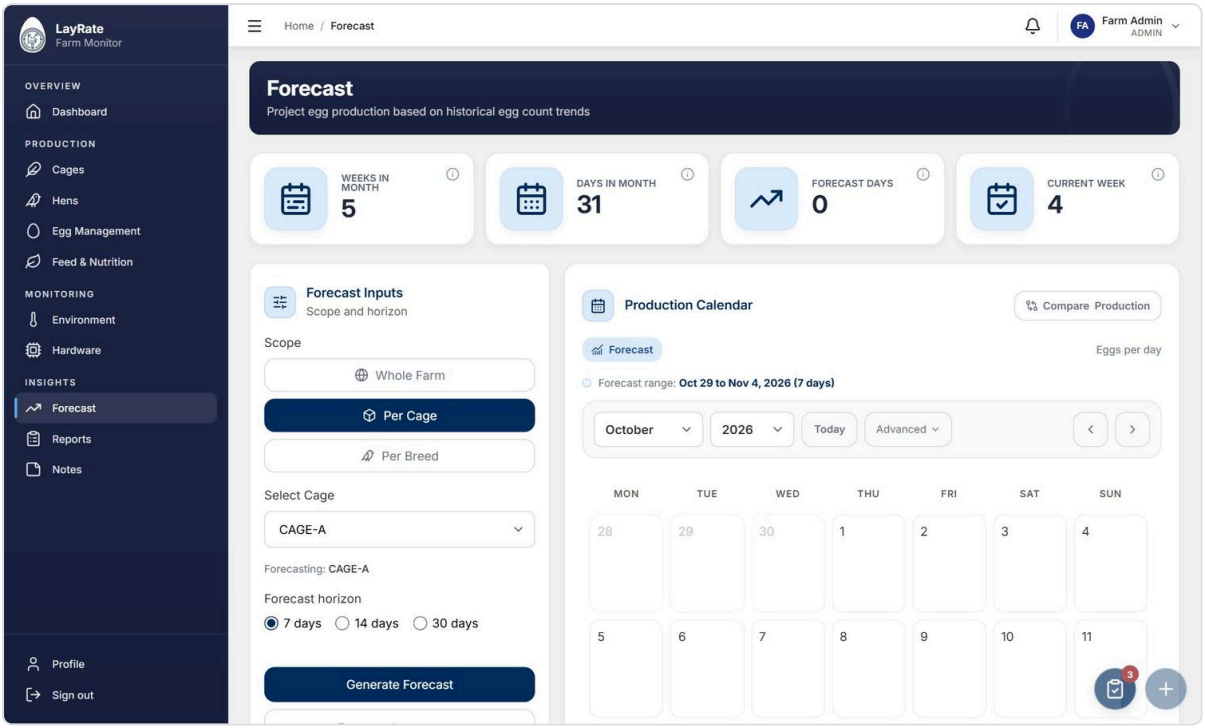


Add Device.

CHAPTER 10

Forecast (Admin)

The Forecast page predicts egg production for the coming days from the farm's history. Forecasts are estimates, not records; they are always shown apart from the actual eggs logged.



The Forecast page.

1. Check the readiness message. A forecast needs at least 90 days of production history for the chosen scope.
2. In **Forecast Inputs**, choose the **Scope**: **Whole Farm**, **Per Cage** (then **Select Cage**) or **Per Breed** (then a breed).
3. Choose the **Forecast horizon**: **7 days**, **14 days** or **30 days**.
4. Tap **Generate Forecast**. The predictions appear on the **Production Calendar**, marked **Forecast**, with the forecast range shown above it. **Reset and regenerate** makes a new one; **Compare Production** compares the forecast with the eggs actually laid.

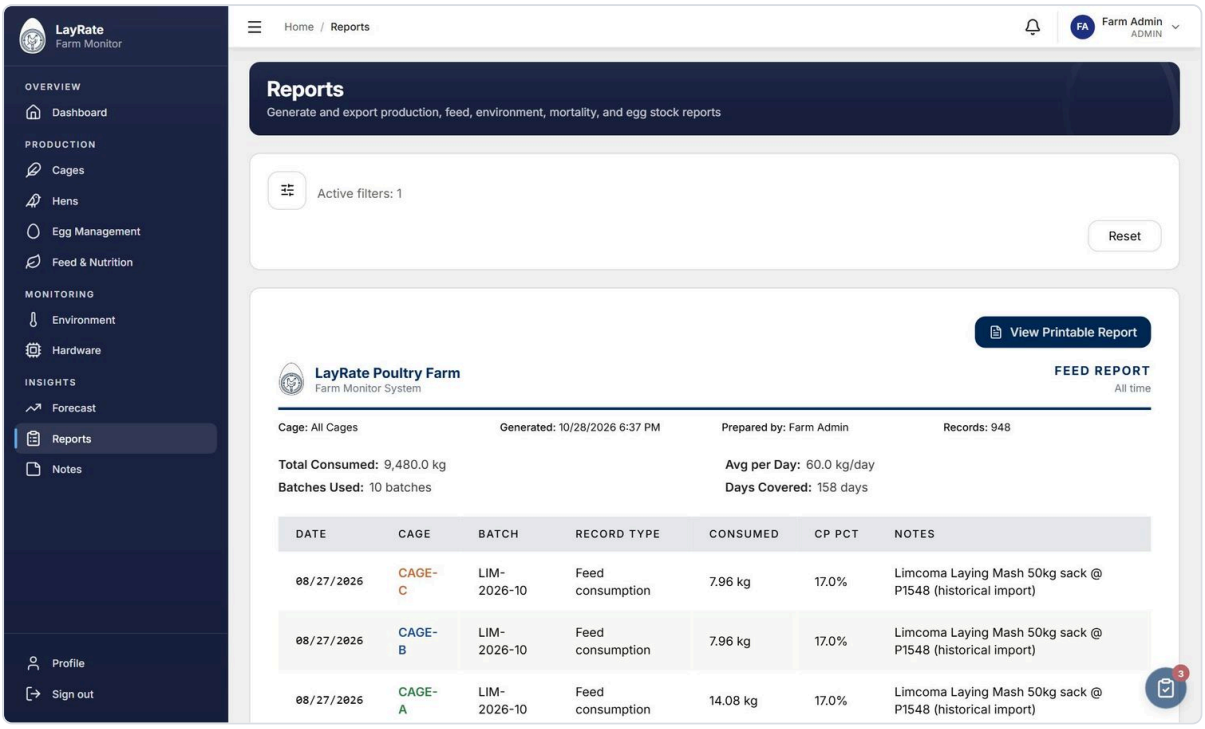
Under **+**:

- **Download input sheet**: an Excel template to fill in with production data.
- **Import production data**: upload the filled-in sheet (.xlsx), check the preview, then confirm.
- **Download Production Data** and **View input records status**: review the data already in LayRate.
- **Export Forecast Report**: save the forecast as PDF, Excel or CSV.

CHAPTER 11

Reports

Reports turns the records into tables you can print or share.



Reports.

1. Choose the report type: **All Reports**, **Production Report**, **Feed Report**, **Environment Report**, **Mortality Report** or **Egg Stock Report**.
2. Tap the filter button to set the dates, cage and the other options shown. Tick **Include graphs** to add charts; for production reports you can tick **Include forecast comparison**.
3. Check the preview: a summary (for example total consumed and average per day) and the table. Tap **Reset** to clear the filters.
4. Tap **Export** and choose **CSV**, **Excel** or **PDF**, or tap **View Printable Report** and print it.

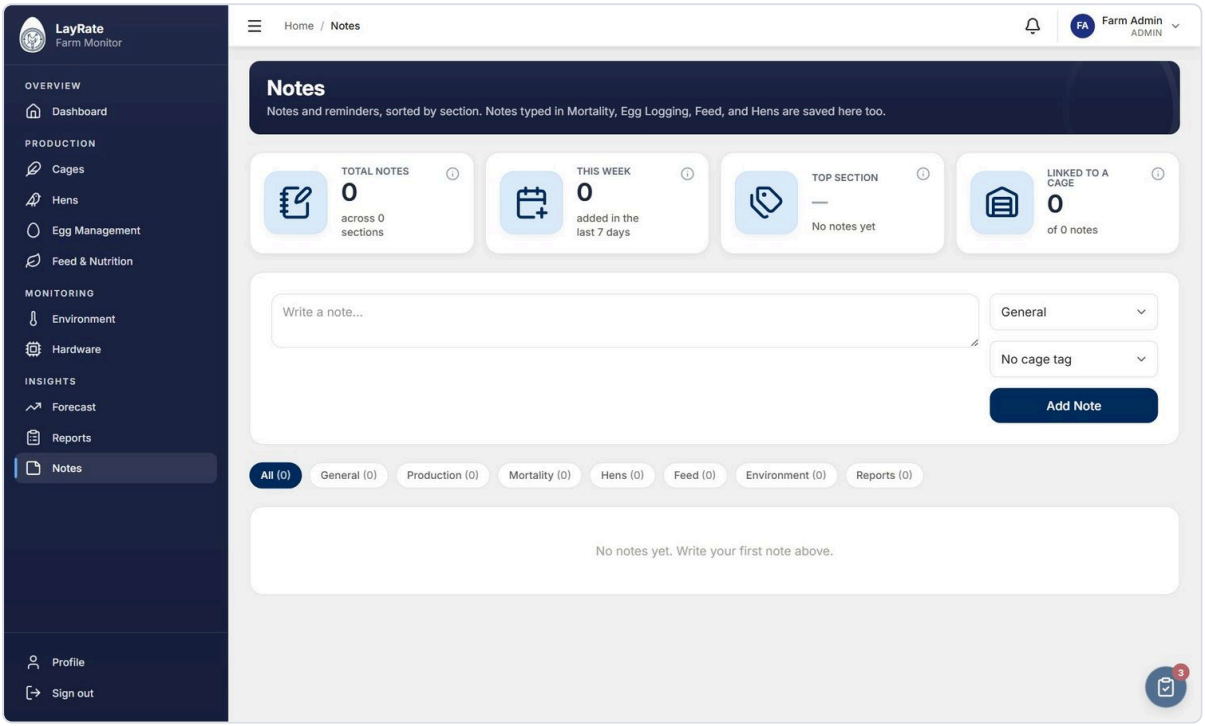
CHAPTER 12

Notes and Notifications

Notes

Notes keeps reminders and observations, sorted by section: General, Production, Mortality, Hens, Feed, Environment and Reports. Notes typed in Mortality, Egg Logging, Feed and Hens are saved here too.

1. Type in the **Write a note...** box.
2. Choose the section (for example **General**) and, if the note is about one cage, the cage tag.
3. Tap **Add Note**. Use the section buttons (**All**, **General**, **Production** ...) to filter the list.



Notes.

Notifications

LayRate warns you about things that need attention, such as temperature or humidity outside the range, low egg or feed stock, and sensor problems. New alerts show a count on the **bell** in the top bar and may pop up when you open a page. Tap the bell to open the **Notifications** page, which lists every alert; tap **Mark all read** (under **+**) when you have checked them. When there are none, it says *No notifications yet*.

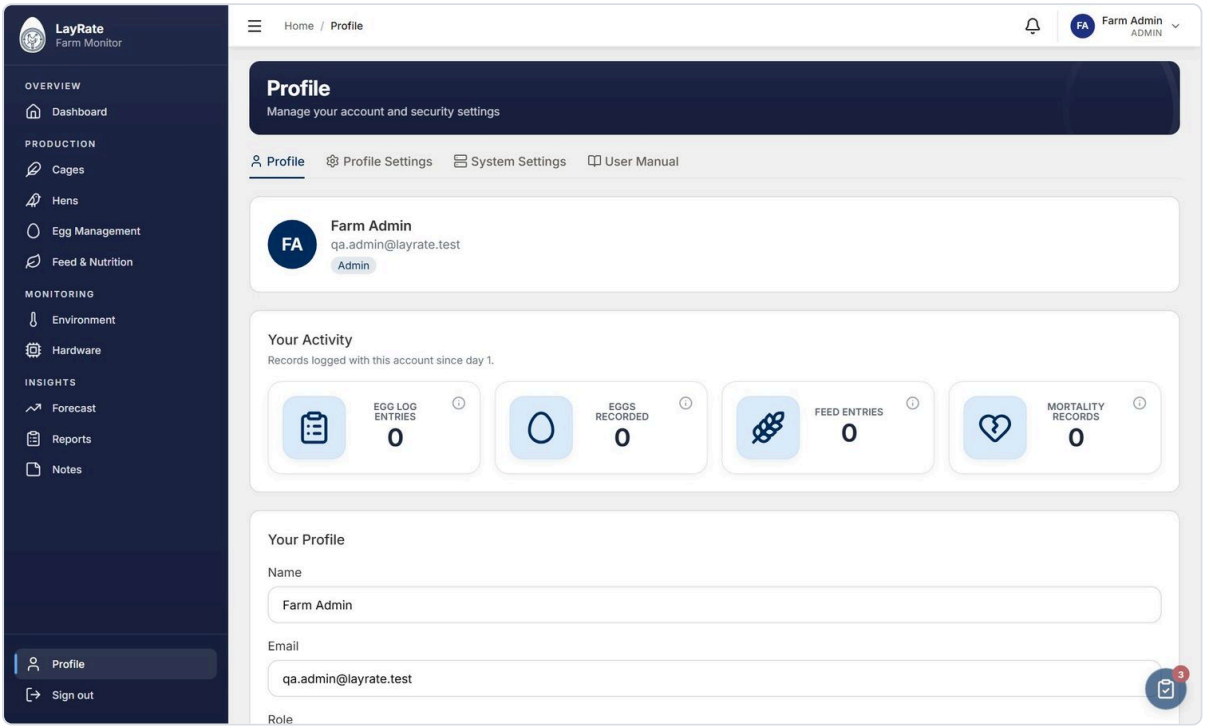
CHAPTER 13

Profile and Settings

Open **Profile** from the bottom of the sidebar or from the menu under your name. It has four tabs.

Profile

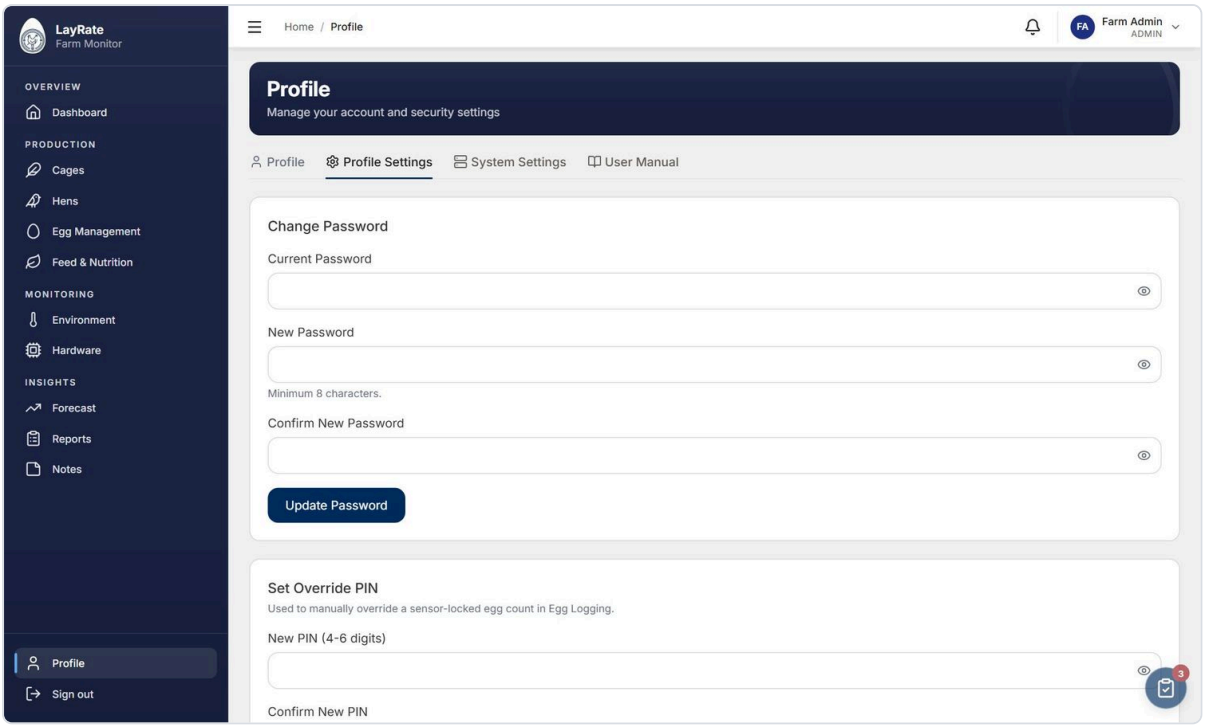
Shows your role, **Your Activity** (how many egg, feed and mortality records you have entered) and **Security Status**. Change your name or email under **Your Profile** and tap **Save Profile**.



Profile.

Profile Settings

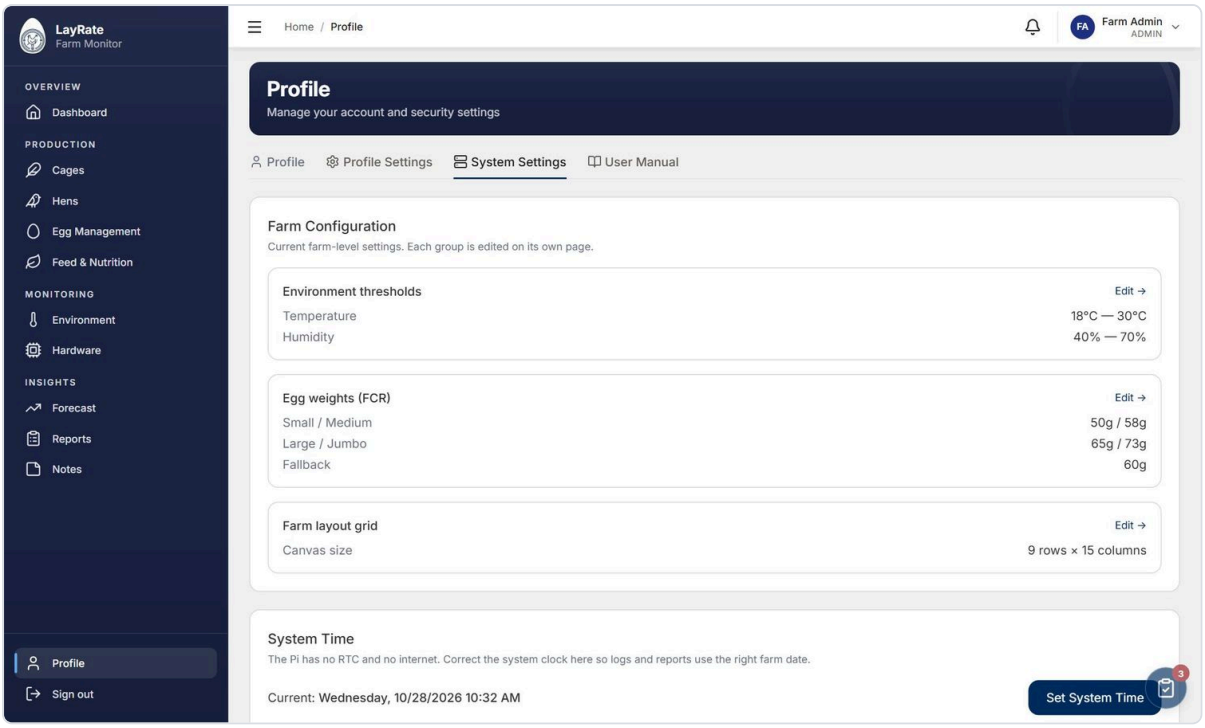
- **Change Password**: enter your **Current Password**, the **New Password** (at least 8 characters) twice, and tap **Update Password**.
- **Set Override PIN**: a 4 to 6 digit code used to correct a sensor egg count in Egg Logging. Weak PINs are refused.
- **Active Sessions**: the devices signed in to your account. **Sign out of all other devices** ends the others.
- **Team** (Admin): add users, edit them, and activate or deactivate accounts. A new user gets a temporary password of at least 8 characters and can change it in their own settings.



Profile Settings.

System Settings (Admin)

- **Farm Configuration**: a summary of the environment thresholds, the egg weights used for FCR, and the farm layout grid size. Tap **Edit** on a group to change it on its own page.
- **System Time**: the farm computer has no internet and no clock battery, so after a power cut its clock can be wrong. Tap **Set System Time** to correct it, so records get the right date.
- **Egg Prices**: price per tray and per piece for each size.
- **Database Backup**: save a copy of all farm data.
- **Clear Database**: permanently deletes all farm data. Make a backup first; this cannot be undone.

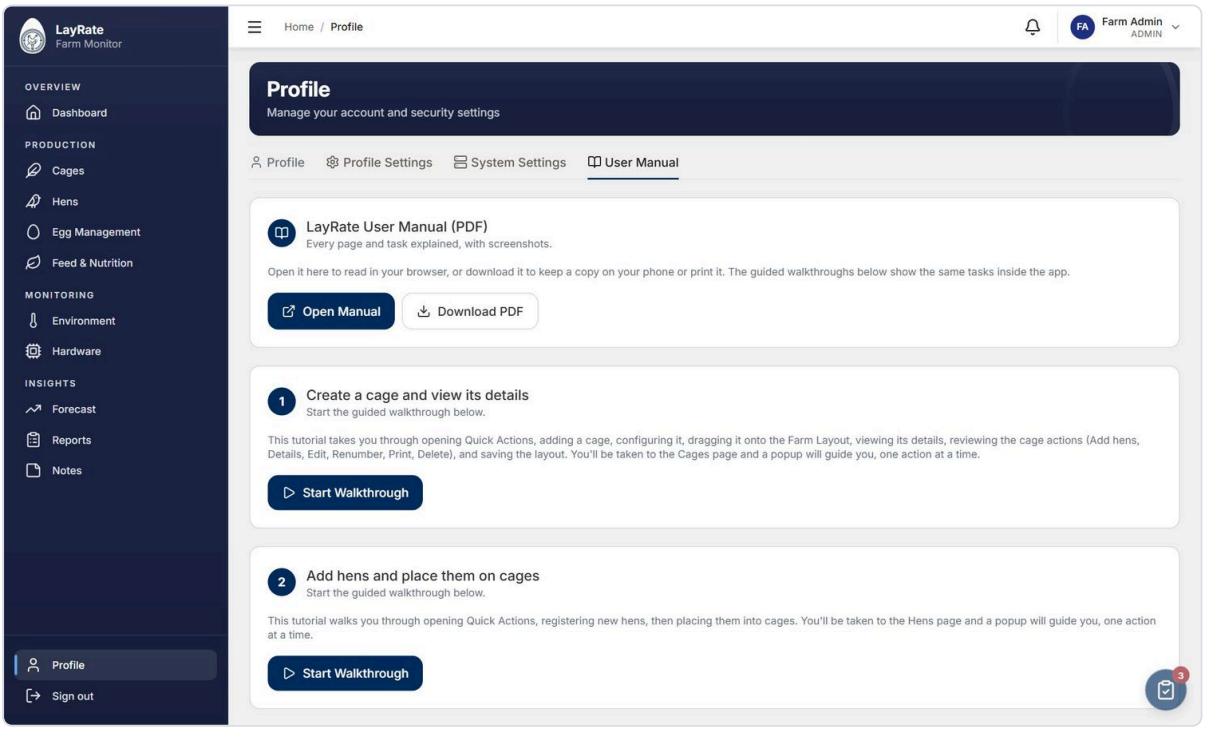


System Settings.

User Manual tab

At the top is this manual as a PDF: **Open Manual** shows it in the browser, and **Download PDF** saves a copy to print or keep on your phone.

Below it are seven guided walkthroughs. Tap **Start Walkthrough** on a task: LayRate opens the right page and a popup guides you one action at a time. The same tasks are written out in the next chapter.



The User Manual tab.

CHAPTER 14

Step-by-Step Tasks

These are the seven walkthroughs from the **User Manual** tab, written out.

Task 1. Create a cage and view its details

1. Open **Cages**. Tap **+** (bottom right), then **Add Cage**.
2. Set **Rows**, **Slots** and **Max/Slot**, then tap **Add Cage**.
3. Drag the new cage from **Unplaced Cages** into the Farm Layout grid. On touchscreens, turn on **Drag cages** first.
4. Tap the **info** button on the placed cage to see its details and actions (Add hens, Details, Edit, Renumber, Print, Delete).
5. Close the details and tap **Save Layout**.

Task 2. Add hens and place them on cages

1. Open **Hens**. Tap **+**, then **Register New Hens**.
2. Set **Quantity**, **Breed**, **Date Acquired** and **Age at Acquisition (weeks)**, then tap **Register Hens**.
3. Tap the **Unplaced** toggle to see the new hens. Use **Next** and **Prev** to page through them.
4. Tap **Place into cage** and assign the hens to a cage.

Task 3. Log daily egg production

1. Open **Egg Management** (Egg Logging). Choose a cage card.
2. Tap a slot, or drag across several slots; the same count applies to each selected slot.
3. Enter the eggs laid for the selected date (0 is valid). For sensor slots, check the reading, or use **Sensor reading – click to override** and verify your PIN or password to change it.
4. For one slot, tap **Save Record**. For several, check the totals and tap **Confirm Save**.

To enter one total for the whole cage instead, use **Total Egg Log** (see Egg Logging).

Task 4. Create feed stock and log daily feeding

1. Open **Feed & Nutrition**. If there is no feed batch yet, tap **+**, then **Add Feed Batch**.
2. Fill in **Brand**, **Crude Protein %**, **Total Quantity (kg)**, **Unit Cost**, **Date Received**, **Low Stock Threshold** and **Notes**, then tap **Add Batch**.
3. Tap **+** again and choose **Add Consumption** (one cage) or **Log Whole-Farm Feeding** (the total, divided among the cages by hen count and marked **Estimated**).
4. Open the **Daily Consumption** tab to see the records you added.

Task 5. Configure the environment and log readings

1. Open **Environment**. Tap **+**, then **Configure Thresholds**.

2. Review **Temp Min / Max (°C)** and **Humidity Min / Max (%)**, then tap **Save Thresholds**.
3. Tap **+**, then **Manual Entry**. Choose a cage or **All Cages**, enter the temperature and humidity, and tap **Save Reading**.
4. Open **Log History** to see the saved readings with their status.

Task 6. Register and assign hardware (Admin)

1. Open **Hardware**. Tap **+**, then **Add Device**.
2. Fill in the device details and tap **Add Device**, then **Add Item** to confirm.
3. Find the new device in the list. Check **Assigned To** (IR sensors can cover more than one slot; spare devices stay unassigned), its **Status**, and **Last Reading** (a new device may have none yet).

Task 7. Forecast egg production (Admin)

1. Open **Forecast** and read the readiness message (90 days of history are needed).
2. Choose **Whole Farm**, **Per Cage** or **Per Breed**, and the number of days to predict.
3. Tap **Generate Forecast** and review the predictions on the production calendar.
4. Use **Export Forecast Report** (under **+**) to save it as PDF, Excel or CSV.

CHAPTER 15

Troubleshooting

Problem	What to do
The LayRate page does not open.	Make sure your device is on the farm Wi-Fi (not mobile data) and that the address is typed correctly. Check that the farm computer is switched on.
I cannot sign in.	See the messages in Signing in. If you forgot your password, ask an Admin to set a new one.
I was sent back to the Sign in page.	Your session ended (you signed out, or another device signed you out). Sign in again.
The Dashboard shows 0 eggs.	No eggs have been logged for that day yet. Log them in Egg Logging, or pick a past date to see earlier records.
A sensor count looks wrong.	Correct it in Egg Logging with **Sensor reading – click to override**, then tell an Admin so the sensor can be checked in Hardware.
Active Sensors dropped, or there are no new readings.	Check the sensor's power and placement. Meanwhile, use **Manual Entry** on the Environment page.
The dates and times look wrong.	The farm computer's clock may have reset after a power cut. An Admin can set it in **System Settings** → **System Time**.
A button I need is missing.	Some actions are for Admins only (see User roles).

CHAPTER 16

Glossary

HDEP	Hen-Day Egg Production. Eggs collected in a day divided by the number of hens, times 100. 90% means 90 eggs from 100 hens.
FCR	Feed Conversion Ratio. Kilograms of feed used per kilogram of eggs. Lower is better.
CP%	Crude protein: the protein percentage on the feed label.
Slot	One compartment of a cage that holds a few hens.
Livability	The share of all hens ever placed that are still alive.
Egg sizes	Small, Medium, Large, XL and Jumbo, set by weight under **Egg Weights**. Unsorted means not yet sized.
Estimated	A feeding row worked out from a whole-farm feeding entry, not measured for that cage.
IR breakbeam sensor	The sensor in a slot that counts eggs as they roll past.
DHT22 sensor	The sensor that reads temperature and humidity.
Relay	The switch that turns the cooling fan on and off.
Override	A sensor count corrected by a person, confirmed with a PIN or password.
Reporting day	The calendar day in Philippine time. A new day starts at midnight.